Quantum Braid Dynamics

A Computational Process

R. Fisher

November 09, 2026

Abstract

Chapter 12 establishes the mathematical convergence of the discrete causal graph sequence to a smooth, four-dimensional pseudo-Riemannian manifold in the thermodynamic limit ($N \to \infty, \ell_0 \to 0$). It addresses the fundamental pathology of coordinate artifacts and geometric instability encountered when continuous spacetime is recovered without a rigorous, measure-sensitive metric. This structural resolution is executed via a dual-limit framework combining spectral geometry and optimal transport. First, the consistently weighted graph Laplacian $\tilde{\mathcal{L}}_t$ is proven to converge in a strong resolvent sense to the continuous Laplace-Beltrami operator $-\Delta_g$, while heat kernel asymptotics provide discrete Li-Yau gradient estimates that guarantee smooth four-dimensional Hausdorff and spectral dimensions. Second, a tensorial averaging map $\mathcal{A}_R$ projects discrete edge scalars onto the unit tangent bundle, where weak convergence to the uniform Haar measure reconstructs smooth symmetric tensor fields like $G_{\mu\nu}$ and $T_{\mu\nu}$. Finally, the directed causal edge distribution breaks local $O(4)$ rotational invariance down to the Lorentz group $SO(3,1)$. This anisotropic drift vector field coupled with a null boundary condition transforms the Riemannian spatial metric into a pseudo-Riemannian metric of Lorentzian signature (-,+,+,+), thereby completing the rigorous synthesis of general relativistic spacetime.

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References **42**
3. **Adams, R. A., & Fournier, J. J. (2003).** 42
29. **Gilbarg, D., & Trudinger, N. S. (2001).** 42
53. **Ollivier, Y. (2009).** 42
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Chapter 12: Continuum Limit (Convergence)

Chapter 12: Continuum Limit (Convergence)

We now ask a critical mathematical question: how does a discrete, relational graph of finite size converge to a smooth, continuous Riemannian manifold in the thermodynamic limit? The previous chapters derived the discrete curvature and field equations, but physical gravity operates on a continuous stage. We must prove that taking the Gromov-Hausdorff-Wasserstein limit of our sequence of graphs reconstructs the smooth kinematics of General Relativity, showing that the discrete relations transition to the continuous fields of classical physics.

Conventional models of quantum gravity often assume a smooth spacetime background from the outset or rely on ad-hoc discretization schemes that break diffeomorphism invariance. Attempts to recover the continuum limit by simply refining a simplex lattice without a rigorous convergence metric lead to coordinate artifacts and structural instabilities, failing to preserve the manifold's dimensionality or smooth structure. Without a spectral convergence mechanism to link the graph Laplacian to the continuous Laplace-Beltrami operator, there is no guarantee that the emergent space will behave like a smooth, **4D** manifold, leaving the continuum limit as an unproven conjecture.

We resolve this mathematical crisis by establishing a rigorous proof of spectral and metric convergence utilizing the tools of Gromov-Hausdorff-Wasserstein geometry. We prove that the spectrum of the discrete graph Laplacian converges to the spectrum of the smooth Laplace-Beltrami operator, and by invoking elliptic regularity and Sobolev embeddings, we guarantee that the limit space is a smooth, **4D** Riemannian manifold. Finally, we construct a **Tensorial Averaging Map** to coarse-grain the discrete edge scalars into smooth, continuous tensor fields, securing a mathematically consistent continuum limit.

Preconditions and Goals

- Prove the Gromov-Hausdorff-Wasserstein Convergence Theorem for the causal graph sequence.
- Establish Laplacian Spectral Convergence to the Laplace-Beltrami operator.
- Formulate the Tensorial Averaging Map to coarse-grain edge scalars.
- Prove that the emergent manifold dimension is exactly 4D using Sobolev embeddings.
- Verify 4D stability against dimensional fluctuations at macroscopic scales.

12.1 Riemannian Convergence

The preceding chapters established that the sequence of causal graphs $\{G_t\}$ at homeostatic equilibrium constitutes a precompact, four-dimensional metric-measure space under the Gromov-Hausdorff-Wasserstein distance. However, a convergent metric-measure space is not intrinsically a smooth manifold; it may possess topological irregularities or lack a differentiable structure. Reconstructing General Relativity requires proving that the discrete graph updates converge to a smooth Riemannian manifold equipped with a well-defined

metric tensor $g_{\mu\nu}$. The central challenge is to demonstrate that discrete operators acting on the graph limit smoothly to the differential operators of continuous geometry, ensuring that scalar fields and causal signals propagate coherently across the emergent spacetime.

Relying solely on metric space convergence is insufficient for physical field theory, as point-set distance convergence does not guarantee the convergence of differential operators or field equations. Discrete graphs can satisfy metric compactness while harboring local spectral pathologies that cause finite-difference operators to diverge or produce unphysical wild oscillations. If the discrete graph Laplacian $\mathcal{L}_t$ fails to converge to the continuum Laplace-Beltrami operator Δ_g, the framework cannot establish differential wave equations or describe smooth scalar field dynamics. A model that lacks spectral operator convergence fails to bridge the gap between combinatorial graph rewrites and the smooth differential geometry of General Relativity.

We resolve this limitation by applying spectral geometry to prove the Laplacian Spectral Convergence Theorem for the causal graph sequence. We demonstrate that the graph Laplacian $\mathcal{L}_t$, properly scaled by the discreteness length ℓ_P, converges in the operator norm to the Laplace-Beltrami operator Δ_g on the limit manifold. Because the spectral properties of the Laplacian uniquely determine the underlying metric tensor, this operator convergence proves that the limit space possesses a smooth, differentiable Riemannian metric. This spectral reconstruction guarantees that the emergent 4-dimensional continuum inherits smooth differential dynamics directly from discrete graph updates.

12.1.1 Definition: Consistently Weighted Laplacian

Specification of the Discrete Laplacian Operator Scaled by the Inverse Square of Discreteness Length

Let $A_{uv} = A_{vu} \geq 0$ be the symmetrized spatial adjacency and $d_u = \sum_v A_{uv}$. At a vertex with $d_u > 0$ set $P_{uv} = A_{uv}/d_u$. At an isolated vertex set $P_{uu} = 1$ and $P_{uv} = 0$ for $v \neq u$. The **Consistently Weighted Laplacian** is

$$\widetilde{\mathcal{L}}_t f(u) = \frac{2(d+2)}{\ell_0^2} \left(f(u) - \sum_v P_{uv} f(v) \right).$$

Here d is the dimension used in the diffusion calibration. On a component of positive degree the inner product is weighted by $\pi_u = d_u / \sum_x d_x$. An isolated vertex is a separate one-dimensional component with zero Laplacian. The physical scaling factor has units of inverse length squared.

12.1.1.1 Commentary: Calibrating Diffusion

Physical Interpretation of the Laplacian Rescaling

To bridge the discrete and the continuum, we must distinguish between the *combinatorics* of a random walk and the *geometry* of diffusion. The standard graph Laplacian $(I - P)$ measures the local variation of a function (its "roughness") but it is dimensionless. It tells us *that* the field is changing, but not *how fast* with respect to physical distance.

The rescaling by ℓ_0^{-2} provides the necessary metric units, converting a finite difference into a second derivative limit ($\partial^2 \sim \Delta f / \Delta x^2$). However, the factor $2(d+2)$ is the crucial physical insight. It accounts for the entropic volume of the step. In 4 dimensions, a random walker has more degrees of freedom to scatter than in 1 dimension. Without this factor, the discrete diffusion process would run at a different "clock rate" than the continuum heat equation requires. This calibration synchronizes the graph diffusion time with the manifold geodesic time, ensuring that the spectral gap encodes the true Ricci curvature of the space rather than an artifact of the lattice dimension.

12.1.2 Theorem: Smooth Manifold Limit

Convergence of the Discrete Causal Graph Sequence to a Smooth Riemannian Manifold via Spectral Convergence

Let $\{G_t\}$ be a refining sequence in the four-dimensional growth regime whose Gromov-Hausdorff limit is identified by a separate comparison with a smooth, compact Riemannian four-manifold (M, g), with the spectral, heat, and elliptic-regularity inputs of the supporting lemmas and a full-rank local spectral coordinate tuple at each point. Then the graph spectral and heat comparisons hold on (M, g), and the limit eigenfunctions supply smooth local charts in which the Laplace-Beltrami principal symbol determines the metric.

12.1.2.1 Commentary: Argument Outline

Structure of the Smooth Riemannian Limit Argument via Spectral Convergence, Heat Kernel Asymptotics, and Smoothness Bootstrapping

The proof proceeds by limits in the four-dimensional geometric growth regime. The spectral, heat, and chart steps use a common smooth comparison and their respective convergence and regularity premises.

```
o 12.1.2 Theorem Smooth Manifold Limit  [by limits]
|
+-- 12.1.3 Lemma: Spectral Convergence
|   +-- 12.1.3.1 Proof: Spectral Convergence
|   +-- 12.1.3.2 Calculation: Spectral Convergence Verification
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|
+-- 12.1.6 Lemma: Ollivier-Ricci Asymptotic Limit
|   +-- 12.1.6.1 Proof: Ollivier-Ricci Asymptotic Limit
|   +-- 12.1.6.2 Commentary: Physical Significance
|
+-- 12.1.7 Proof: Smooth Manifold Limit
```

12.1.3 Lemma: Spectral Convergence

Asymptotic Convergence of the Discrete Spectrum to the Continuum Laplace-Beltrami Eigenvalues via Spectral Convergence

Given the conditions of **Eigenvalues** and **Eigenfunctions**, the properties of Asymptotic Convergence of the Discrete Spectrum to the Continuum Laplace-Beltrami Eigenvalues are established.

12.1.3.1 Proof: Spectral Convergence

Operator Decomposition via Perturbation Analysis

I. Reversibility and the energy form

On a positive-degree component the spatial walk satisfies

$$\pi_u P_{uv} = \frac{A_{uv}}{\sum_x d_x} = \frac{A_{vu}}{\sum_x d_x} = \pi_v P_{vu}$$

(**Consistently Weighted Laplacian**). Thus $\sum_u \pi_u P_{uv} = \pi_v$. For real functions f, g,

$$\begin{aligned}\frac{1}{2}\sum_{u,v} \pi_u P_{uv}(f(u) - f(v))(g(u) - g(v)) &= \sum_u \pi_u f(u) g(u) - \sum_{u,v} \pi_u P_{uv} f(u) g(v) \\ &= \langle f, (I - P) g \rangle_\pi.\end{aligned}$$

The form is symmetric and nonnegative on the diagonal. On a regular grid of degree q, $I - P = (D - A)/q$. The unnormalized grid benchmark therefore uses a different constant calibration (**Spectral Convergence Verification**).

II. Pointwise error and bandwidth

With discrete and continuum functions compared in a common norm, the fixed-test-function error decomposes as

$$\|\widetilde{\mathcal{L}}_t f + \Delta_g f\| \le \|\mathbb{E}[\widetilde{\mathcal{L}}_t] f + \Delta_g f\| + \|(\widetilde{\mathcal{L}}_t - \mathbb{E}[\widetilde{\mathcal{L}}_t]) f\|.$$

A pointwise estimate depends on the bandwidth of its own stencil. In particular, if the bandwidth in the fluctuation denominator is ℓ_0 and $N_t \asymp \ell_0^{-4}$, then

$$\sqrt{N_t \ell_0^4} \asymp \sqrt{\ell_0^{-4} \ell_0^4} = 1.$$

The pointwise bound with this denominator gives no $N_t^{-1/2}$ decay.

III. Projection onto a mode

For a continuum eigenfunction f_k, a global mode projection can be written as $N_t^{-1} \sum_i Y_i$. Its variance is

$$\mathrm{Var}\left(\frac{1}{N_t} \sum_i Y_i \right) = \frac{1}{N_t^2} \sum_{i,j} \mathrm{Cov}(Y_i, Y_j).$$

A bound of order N_t on the covariance sum of the scaled terms would give variance of order N_t^{-1}. This is a separate condition on the mode projection. The covariance estimate concerns a single mode.

IV. Balanced-stencil and mesoscopic comparisons

Write $x_v - x_u = \ell_0 z_{uv}$ in local normal coordinates at x_u, with bounded stencil vectors, and define $m_u = \sum_v P_{uv} z_{uv}$ and $Q_u = \sum_v P_{uv} z_{uv} \otimes z_{uv}$. For a smooth fixed test f, Taylor expansion gives

$$\begin{aligned} f(x_v) &= f(x_u) + \ell_0 \nabla f(x_u) \cdot z_{uv} + \frac{\ell_0^2}{2} \nabla^2 f(x_u) : (z_{uv} \otimes z_{uv}) + O_f(\ell_0^3), \\ \ell_0^{-2}(I - P) f(u) &= -\ell_0^{-1} \nabla f(x_u) \cdot m_u - \tfrac{1}{2} \nabla^2 f(x_u) : Q_u + O_f(\ell_0). \end{aligned}$$

The first term vanishes on a balanced stencil with $m_u = 0$. In the reversible energy form each difference is of order ℓ_0. For two smooth tests f, g the leading product is

$$\begin{aligned} \ell_0^{-2}(f(u) - f(v))(g(u) - g(v)) &= (\nabla f(x_u) \cdot z_{uv})(\nabla g(x_u) \cdot z_{uv}) + O_{f,g}(\ell_0), \\ \ell_0^{-2} \langle f, (I - P) g \rangle_\pi &= \tfrac{1}{2} \sum_u \pi_u \, \nabla f(x_u)^{\mathsf{T}} Q_u \nabla g(x_u) + O_{f,g}(\ell_0). \end{aligned}$$

The second line uses $\sum_{u,v} \pi_u P_{uv} = 1$. For a target second-moment field $Q(x)$ and a continuum measure ν, put $H_{f,g}(x) = \nabla f(x)^{\mathsf{T}} Q(x) \nabla g(x)$. The weak comparison for fixed tests satisfies

$$\left| \ell_0^{-2} \langle f, (I-P)g \rangle_\pi - \tfrac{1}{2} \int H_{f,g} \, \mathrm{d}\nu \right| \le C_{f,g} \ell_0 + \tfrac{1}{2} \|\nabla f\|_\infty \|\nabla g\|_\infty \sum_u \pi_u \|Q_u - Q(x_u)\| + \tfrac{1}{2} \left| \sum_u \pi_u H_{f,g}(x_u) - \int H_{f,g} \, \mathrm{d}\nu \right|.$$

The last two terms come from replacing Q_u with $Q(x_u)$ at each vertex and then replacing the discrete measure by ν. In the Laplace-Beltrami comparison the normalization of Q and ν must give the intended continuum Dirichlet form. A variational resolvent comparison additionally needs interpolation, control of varying sequences of tests, and energy compactness.

In the mesoscopic kernel route one instead compares an operator built with bandwidth h_t, which is distinct from the one-hop spacing. For example, the ε-graph comparison of (Calder & Garcia Trillos, 2022) uses independent samples with a smooth positive density bounded away from zero on a compact smooth manifold without boundary, and the small-bandwidth sampling regime $(\log N_t/N_t)^{1/d} \ll h_t \ll 1$ for eigenvalue consistency. For the sampled points x_i the operator and its continuum normalization are

$$(L_{h_t} f)(x_i) = \frac{1}{N_t h_t^{d+2}} \sum_{j=1}^{N_t} \eta\left(\frac{|x_i - x_j|}{h_t} \right) (f(x_i) - f(x_j)),$$
$$A_{\rho,\eta} f = -\frac{\sigma_\eta}{2\rho} \operatorname{div}_g(\rho^2 \nabla_g f), \qquad \sigma_\eta = \int_{\mathbb{R}^d} z_1^2 \eta(|z|) \, \mathrm{d}z.$$

Here ρ is the sampling density. The nonnegative radial kernel profile is nonincreasing, supported on $[0,1]$, Lipschitz on that interval, and positive at $1/2$, with radial integral one. For a constant ρ, $A_{\rho,\eta} = -(\sigma_\eta \rho/2)\Delta_g$, so the comparison needs a constant calibration even in that case. Applying this route to the causal graph would require identifying its sampling, weights, and limit operator with those of the comparison.

Powered lazy walk.

Let P_t be reversible and lazy with spectrum in $[0,1]$, and define $A_t = a_t(I - P_t)$ and $A_{t,k} = a_t(I - P_t^k)/k$ with $a_t > 0$ and integer $k \ge 1$. If $A_t u = \lambda u$, then $P_t u = (1 - \lambda/a_t)u$, so the powered eigenvalue on u is $F_t(\lambda) = a_t[1 - (1 - \lambda/a_t)^k]/k$. The map increases strictly on $[0, a_t]$. For $x = \lambda/a_t$,

$$0 \le \lambda - F_t(\lambda) = a_t \int_0^x [1 - (1-y)^{k-1}] \, \mathrm{d}y \le a_t(k-1) \int_0^x y \, \mathrm{d}y = \frac{(k-1)\lambda^2}{2a_t}.$$

Conversely, for $b = kF_t(\lambda)/a_t < 1$, inverting the power gives

$$\lambda = a_t[1 - (1-b)^{1/k}] = \frac{a_t}{k} \int_0^b (1-y)^{1/k - 1} \, \mathrm{d}y \le \frac{F_t(\lambda)}{1-b}.$$

If $k/a_t \to 0$ and the powered eigenvalues in a fixed band are bounded, then $b \to 0$ uniformly there. The inverse estimate bounds the associated one-hop eigenvalues, and the first estimate gives a difference tending to zero. The eigenspaces are the same because the map is strictly increasing. This transfers a fixed-band spectral comparison already proved for $A_{t,k}$.

V. Eigenvalues on a common Hilbert space

Let $A = -\Delta_g \geq 0$ be self-adjoint with compact resolvent on $H = L^2(M, \mathrm{d}V_g)$, and let J_t be an isometric comparison map from $L^2(\pi)$ to H. Take $\widetilde{\mathcal{L}}_t$ to be the nonnegative, self-adjoint graph operator. The spectral comparison uses the following norm-resolvent hypothesis:

$$R = (I + A)^{-1}, \qquad R_t = J_t(I + \widetilde{\mathcal{L}}_t)^{-1}J_t^*, \qquad \varepsilon_t = \|R_t - R\|_{H\to H} \longrightarrow 0.$$

For instance, in a weak form comparison, suppose the embedded resolvents converge strongly to R and the union of their images of the unit ball is relatively compact. These properties give the norm bound in the limit. If norm convergence fails, take unit vectors b_t along a subsequence with $\|(R_t - R)b_t\| \geq c > 0$. Pass to subsequences with $b_t \rightharpoonup b$ and $R_t b_t \to y$. For each fixed $g \in H$, self-adjointness gives

$$\langle y, g\rangle = \lim_t \langle b_t, R_t g\rangle = \langle b, Rg\rangle = \langle Rb, g\rangle.$$

Then $y = Rb$, and compactness of R gives $Rb_t \to Rb$, contradicting the choice of b_t. A quantitative bound on ε_t would require a quantitative comparison.

Here R_t acts as zero on the orthogonal complement of the embedded space. Write $r_k = (1 + \lambda_k)^{-1}$ and $r_{t,k} = (1 + \widetilde{\lambda}_{t,k})^{-1}$ for the positive eigenvalues of the two resolvents in descending order. For any unit $u \in H$,

$$\langle u, Ru\rangle - \varepsilon_t \leq \langle u, R_t u\rangle \leq \langle u, Ru\rangle + \varepsilon_t.$$

Taking the infimum over unit vectors in a k-dimensional subspace L and then the supremum over L gives, with indexing starting at $k = 1$,

$$r_k - \varepsilon_t \leq \sup_{\dim L = k} \inf_{\substack{u\in L\\ \|u\|=1}} \langle u, R_t u\rangle = r_{t,k} \leq r_k + \varepsilon_t.$$

For $\varepsilon_t < r_k/2$ the kth graph resolvent eigenvalue is positive and $r_{t,k} \geq r_k/2$. Hence

$$\left|\widetilde{\lambda}_{t,k} - \lambda_k\right| = \left|\frac{1}{r_{t,k}} - \frac{1}{r_k}\right| = \frac{|r_{t,k} - r_k|}{r_{t,k} r_k} \leq \frac{2\varepsilon_t}{r_k^2}.$$

VI. Eigenfunctions and isolated eigenspaces

For an isolated simple eigenvalue $r = r_k$ of R, let f_k be its unit eigenfunction, P_k the orthogonal projection onto its span, and $\gamma = \operatorname{dist}(r, \sigma(R) \setminus \{r\}) > 0$. A unit embedded graph eigenfunction $u_t = J_t\psi_{t,k}$ satisfies

$$(R - rI)u_t = (R - R_t)u_t + (r_{t,k} - r)u_t, \qquad \|(R - rI)u_t\| \leq 2\varepsilon_t.$$

Expansion in the orthonormal eigenbasis of R gives the gap estimate

$$\gamma^2\|(I - P_k)u_t\|^2 = \gamma^2 \sum_{j\neq k} |\langle f_j, u_t\rangle|^2 \leq \sum_{j\neq k} |r_j - r|^2 |\langle f_j, u_t\rangle|^2 = \|(R - rI)u_t\|^2.$$

Choose the phase of $\psi_{t,k}$ so that $\langle f_k, u_t\rangle$ is nonnegative. Since $J_t^* J_t = I$, the graph comparison with the pulled-back eigenfunction obeys

$$\begin{aligned}\|\psi_{t,k} - J_t^* f_k\|_{L^2(\pi)}^2 &\leq \|u_t - f_k\|_H^2 = 2(1 - |\langle f_k, u_t\rangle|)\\ &\leq 2(1 - |\langle f_k, u_t\rangle|^2) = 2\|(I - P_k)u_t\|^2 \leq \frac{8\varepsilon_t^2}{\gamma^2}.\end{aligned}$$

For a repeated isolated eigenvalue r, let P and P_t project onto the spectral clusters inside the circle $\mathcal{C} = \{z : |z - r| = \gamma/2\}$. When $\varepsilon_t < \gamma/4$, on $\mathcal{C}$ the two resolvent norms satisfy $\|(zI - R)^{-1}\| \le 2/\gamma$ and $\|(zI - R_t)^{-1}\| \le 4/\gamma$. The second bound follows by factoring $zI - R_t = (zI - R)[I - (zI - R)^{-1}(R_t - R)]$ and bounding the inverse of the second factor by the geometric series. The contour projection formula and the resolvent identity give

$$P_t - P = \frac{1}{2\pi i} \oint_{\mathcal{C}} (zI - R_t)^{-1}(R_t - R)(zI - R)^{-1}\, \mathrm{d}z,$$
$$\|P_t - P\| \le \frac{\pi\gamma}{2\pi} \frac{4}{\gamma} \varepsilon_t \frac{2}{\gamma} = \frac{4\varepsilon_t}{\gamma}.$$

Thus each fixed eigenvalue converges, simple eigenfunctions converge after phase alignment, and repeated isolated eigenvalues give convergence of the associated eigenspaces under the common-space hypothesis.

Q.E.D.

12.1.3.2 Calculation: Spectral Convergence Verification

Verification of Laplacian Spectral Convergence via Periodic 4D Grid Approximations

The following protocols check eigenvalue convergence on the periodic grid benchmark (**Spectral Convergence**). The grid normalization is related to the reversible spatial walk by the degree factor on a regular graph (**Consistently Weighted Laplacian**).

1. **Grid Discretization:** The algorithm constructs a sequence of periodic 4D grid graphs representing discrete approximations of the Riemannian manifold.
2. **Spectrum Eigendecomposition:** The protocol performs numerical eigendecomposition of the consistently weighted discrete Laplacian to isolate the first non-zero eigenvalue.
3. **Convergence Scaling Check:** The metric tracks the convergence of the discrete eigenvalue toward the analytical Laplace-Beltrami target to validate the expected second-order error scaling.

```
import numpy as np
import networkx as nx
from scipy.sparse.linalg import eigsh
from scipy.sparse import diags
from itertools import product

def toy_4d_grid(N):
    """
    Constructs a periodic 4D grid graph (Torus) with N nodes.
    Ensures Ahlfors 4-regularity by construction.
    """
    k = int(round(N**(1/4)))
    if k**4 != N:
        raise ValueError(f"N={N} is not a perfect 4th power.")

    dim = [k] * 4
    G = nx.grid_graph(dim=dim, periodic=True)

    # Flatten node labels for matrix operations
    mapping = {tuple(idx): i for i, idx in enumerate(product(range(k), repeat=4))}
    G = nx.relabel_nodes(G, mapping)
    return G, 1.0/k  # Graph and fundamental scale ell_0

def compute_fiedler_value(G, ell0):
    """
```

```
    Computes the first non-zero eigenvalue of the Rescaled Laplacian.
    L_tilde = (1/ell0^2) * (D - A) [Unnormalized form matches grid geometry]
    """
    A = nx.adjacency_matrix(G).astype(float)
    degrees = np.array(A.sum(axis=1)).flatten()

    # Construct Unnormalized Laplacian L = D - A
    # Unnormalized form is used because on a regular grid D is constant (2d),
    # matching the standard finite difference Laplacian.
    L_unnorm = diags(degrees) - A

    # Apply Metric Scaling: 1 / ell_0^2
    factor = 1.0 / (ell0**2)
    L_scaled = factor * L_unnorm

    # Solve for k=6 smallest magnitude eigenvalues
    # Shift-invert mode would be faster, but SM with sort is robust here.
    try:
        vals = eigsh(L_scaled, k=6, which='SM', return_eigenvectors=False)
        vals = np.sort(vals)

        # Filter numerical zeros (machine precision)
        non_zeros = vals[vals > 1e-5]

        if len(non_zeros) > 0:
            return non_zeros[0] # The Fiedler value
        else:
            return 0.0
    except Exception as e:
        return np.nan

print("--- Spectral Convergence Verification (4D Torus) ---")
print("Target Continuum Eigenvalue: (2*pi)^2 ~= 39.4784")
print(f"{'N':<8} | {'ell_0':<8} | {'Lambda_1':<10} | {'Theory':<10} | {'Error %':<10}")
print("-" * 60)

target = (2 * np.pi)**2

for k in [4, 6, 8, 10]:
    N = k**4
    G, ell0 = toy_4d_grid(N)
    lam = compute_fiedler_value(G, ell0)
    err = abs(lam - target) / target * 100
    print(f"{N:<8} | {ell0:<8.4f} | {lam:<10.4f} | {target:<10.4f} | {err:<10.2f}")
```

The fixture uses the unnormalized grid stencil $D - A$ with its own calibration. On a regular grid of degree q, this stencil equals $q(I - P)$ for the reversible spatial walk.

Simulation Results:

```
--- Spectral Convergence Verification (4D Torus) ---
Target Continuum Eigenvalue: (2*pi)^2 ~= 39.4784
N        | ell_0    | Lambda_1   | Theory     | Error %
------------------------------------------------------------
256      | 0.2500   | 32.0000    | 39.4784    | 18.94
```

1296	0.1667	36.0000	39.4784	8.81
4096	0.1250	37.4903	39.4784	5.04
10000	0.1000	38.1966	39.4784	3.25

Conclusion: The simulation confirms the spectral convergence of the discrete Laplacian to the continuum limit. The first non-zero eigenvalue λ_1 approaches the theoretical value of $(2\pi)^2 \approx 39.48$ as the graph resolution refines ($\ell_0 \to 0$). The error scales monotonically with the edge length, consistent with the expected discretization error of the operator on a regular lattice. The benchmark uses the regular-grid operator and its grid calibration. The general comparison still requires the operator hypotheses of the proof.

12.1.3.3 Commentary: Hearing the Shape of Spacetime

Interpretation of Spectral Convergence via Geometric Invariants

Establishing the spectral convergence of the discrete graph Laplacian to the continuous Laplace-Beltrami operator provides a mathematical resolution to the discrete analogue of Mark Kac's classical question: "Can one hear the shape of a drum?" In relational quantum gravity, the causal graph serves as the physical substrate, while the discrete Laplacian spectrum $\tilde{\lambda}_k$ represents the fundamental vibrational frequencies of spacetime geometry.

Proving that the discrete spectrum limits monotonically to the continuous manifold spectrum ($\tilde{\lambda}_k \to \lambda_k$ as $\ell_0 \to 0$) demonstrates a deep structural equivalence between the discrete graph and continuous Riemannian manifolds. The Laplacian eigenvalues λ_k encode coordinate-invariant geometric data, including spatial volume, topological dimension, scalar curvature, and global Betti numbers. Spectral convergence guarantees that relational graph dynamics preserve intrinsic geometric invariants without relying on ad-hoc coordinate embeddings.

This spectral correspondence ensures that physical wave equations and quantum field propagators defined on the discrete graph faithfully reproduce continuum physics in the macroscopic limit. Because the discrete Laplacian spectrum converges to the smooth manifold spectrum, field fluctuations on the graph experience the exact metric curvature and topological boundary conditions of the continuum manifold. The relational causal graph does not merely mimic smooth geometry; it possesses the exact spectral resonance of continuous spacetime.

12.1.4 Lemma: Heat Kernel Asymptotics

Demonstration of Gaussian Heat Kernel Bounds via Discrete Li-Yau Estimates

Suppose $p_t(x, y)$ is the heat kernel on the causal graph G_t. Then it converges asymptotically to the Gaussian fundamental solution of the continuum heat equation.

12.1.4.1 Proof: Heat Kernel Asymptotics

Derivation of Heat Kernel Bounds from Functional Inequalities on the Graph

I. Reversible walk and volume doubling

Consider the spatial walk of **Consistently Weighted Laplacian** with reversible measure μ. In the calibrated scale window, assume $c_1 s^4 \le V(x, s) \le c_2 s^4$ for $s = r, 2r$ (**Conditional Ball Growth**). Then

$$V(x, 2r) \le c_2(2r)^4 = 16c_2 r^4 \le 16(c_2/c_1)V(x, r).$$

II. Functional inequality for a Gaussian comparison

Let $c_{uv} = \mu(u)P_{uv} = c_{vu}$ and let f_{B_r} be the μ-average over $B_r(x)$. The local Poincare input on the comparison window is a bound of the form

$$\sum_{u\in B_r}\mu(u)|f(u)-f_{B_r}|^2\le C_P r^2\sum_{\{u,v\}\subset B_{\eta r}}c_{uv}|f(u)-f(v)|^2,\qquad \eta\ge 1.$$

Here C_P and the allowed enlargement factor η are uniform, and the balls used in the inequality lie in the comparison regime. In a graph Gaussian estimate based on doubling and Poincare, one must supply the local inequality and the transition hypotheses of that estimate, including a uniform lower bound on positive transition probabilities and laziness in the discrete-time formulation. For that formulation the spatial walk can be made lazy by replacing P by $(I+P)/2$, which preserves the reversible measure and halves $I-P$. With the applicable comparison hypotheses, the Gaussian bounds control the kernel up to constants in the exponent and the volume prefactor. In the diffusive range they take the form

$$\frac{c}{V(x,\sqrt{s})}e^{-Cd(x,y)^2/s}\le p_s(x,y)\le\frac{C'}{V(x,\sqrt{s})}e^{-c'd(x,y)^2/s}.$$

Here p_s is a density relative to μ, and the constants and range are those of the selected comparison theorem.

III. Alternative curvature-dimension route

A graph Li-Yau estimate uses a curvature-dimension hypothesis for the chosen generator L. In the exponential curvature-dimension formulation, write $\Gamma(f,g)=\frac{1}{2}[L(fg)-fLg-gLf]$, $\Gamma(f)=\Gamma(f,f)$, and $\Gamma_2(f)=\frac{1}{2}L\Gamma(f)-\Gamma(f,Lf)$. The condition $\mathrm{CDE}(n,-K)$ requires at vertices with $f>0$ and $Lf<0$ that

$$\Gamma_2(f)-\Gamma(f,\Gamma(f)/f)\ge\frac{(Lf)^2}{n}-K\Gamma(f).$$

The condition concerns the reversible generator and its iterated energy form. The retained-edge curvature result in **Curvature Monotonicity** concerns a transport comparison for a specific graph update. A graph Li-Yau route requires verification of its own generator hypothesis.

IV. Continuum kernel and the small-time coefficient

Use a common-space heat-kernel comparison to identify the limit kernel with the heat kernel of $-\Delta_g$ for an independently smooth Riemannian metric. For continuum diffusion time $\tau>0$, the smooth diagonal parametrix gives

$$p_\tau^g(x,x)=(4\pi\tau)^{-2}\left(1+\tfrac{\tau}{6}R_g(x)+O(\tau^2)\right).$$

Away from the diagonal the local parametrix includes a separate leading transport coefficient multiplying the Gaussian factor. To transfer the diagonal curvature coefficient along a joint limit $\tau=\tau_t\to 0$, the volume-normalized kernel comparison would need to control the error on that scale. In particular, if the embedded density $p_\tau^{(t)}$ satisfies

$$(4\pi\tau_t)^2\left(p_{\tau_t}^{(t)}(x,x)-p_{\tau_t}^g(x,x)\right)=o(\tau_t),$$

then

$$\frac{(4\pi\tau_t)^2p_{\tau_t}^{(t)}(x,x)-1}{\tau_t}=\frac{R_g(x)}{6}+O(\tau_t)+o(1).$$

V. Fixed-cutoff heat trace and order of limits

Let A_t and A be the nonnegative graph and comparison generators with eigenvalues $\lambda_{t,j}$ and λ_j in increasing order with multiplicity. At fixed t, the spectrum is finite and

$$\lim_{s\downarrow 0}\operatorname{Tr}(e^{-sA_t}) = \sum_{j=1}^{N_t}\lim_{s\downarrow 0} e^{-s\lambda_{t,j}} = N_t.$$

Now fix a physical diffusion time $s > 0$ and a positive cutoff Λ outside the spectrum of A. Let m count the eigenvalues below Λ. The fixed-index convergence from the operator comparison (**Spectral Convergence**) puts $\lambda_{t,m} < \Lambda < \lambda_{t,m+1}$ eventually if $m \geq 1$. For $m = 0$ it puts $\lambda_{t,1} > \Lambda$. For nonnegative a, b, integration of the exponential derivative gives $|e^{-sa} - e^{-sb}| \leq s|a - b|$. Thus

$$\left|\sum_{\lambda_{t,j}<\Lambda} e^{-s\lambda_{t,j}} - \sum_{\lambda_j<\Lambda} e^{-s\lambda_j}\right| \leq s\sum_{j=1}^{m}|\lambda_{t,j} - \lambda_j| \longrightarrow 0.$$

If A is the Laplacian of a smooth compact four-dimensional Riemannian manifold without boundary, e^{-sA} is trace class. At fixed s, taking $\Lambda \to \infty$ after the graph limit recovers its trace. Integrating the continuum diagonal expansion then gives

$$\operatorname{Tr}(e^{-sA}) = \int_M p_s^g(x,x)\,\mathrm{d}V_g = (4\pi s)^{-2}\left(\operatorname{Vol}_g(M) + \frac{s}{6}\int_M R_g\,\mathrm{d}V_g + O(s^2)\right) \quad (s \downarrow 0).$$

An untruncated discrete trace limit at fixed s also needs a uniform heat-tail bound. Taking s to zero together with the mesh requires control of that comparison on the chosen diffusion window.

VI. Removing the cutoff at fixed diffusion time

Fix $s > 0$. Suppose the calibrated graph kernel at time $s/2$ has a uniform diagonal bound $p^{(t)}_{s/2}(x,x) \leq M_s$ with respect to the reversible measure μ_t, and suppose $\mu_t(V_t) \leq M_0$. For the orthonormal basis of normalized point indicators in $L^2(\mu_t)$, the diagonal entry of $e^{-(s/2)A_t}$ at x is $\mu_t(x)p^{(t)}_{s/2}(x,x)$. Therefore

$$\operatorname{Tr} e^{-(s/2)A_t} = \sum_{x\in V_t}\mu_t(x)p^{(t)}_{s/2}(x,x) \leq M_0M_s,$$
$$\sum_{\lambda_{t,j}>\Lambda} e^{-s\lambda_{t,j}} \leq e^{-s\Lambda/2}\sum_{\lambda_{t,j}>\Lambda} e^{-(s/2)\lambda_{t,j}} \leq M_0M_se^{-s\Lambda/2}.$$

At fixed s, split the difference of the full graph and continuum traces into the two tails and the fixed-band difference of Step V. First take the graph limit, then $\Lambda \to \infty$. The displayed bound controls the graph tail and trace class controls the continuum tail. A uniform version on a shrinking diffusion window needs additional control of M_s and of the kernel comparison on that window.

Q.E.D.

12.1.4.2 Calculation: Heat Kernel Asymptotics Verification

Validation of Heat Kernel Asymptotics via Matrix Exponential Diffusion Solvers

Verification of the short-time Gaussian diffusion **Heat Kernel Asymptotics** established by **Gaussian Bounds** and **Heat Kernel Asymptotics** is based on the following protocols:

1. **Heat Kernel Computation:** The algorithm computes the recurrence probability at a reference node using the matrix exponential of the discrete Laplacian.
2. **Dimensional Extraction:** The protocol evaluates the slope of the recurrence probability in the short-time logarithmic regime to estimate the effective system dimension.
3. **Resolution Convergence Analysis:** The metric tracks the convergence of the effective dimension toward the target value as the grid resolution increases.

```python
import numpy as np
import networkx as nx
from scipy.optimize import curve_fit
from itertools import product
from scipy.sparse.linalg import expm_multiply
from scipy.sparse import eye, diags

def toy_4d_grid(N):
    k = int(round(N**(1/4)))
    if k**4 != N:
        raise ValueError("N must be k^4")
    dim = [k] * 4
    G = nx.grid_graph(dim=dim, periodic=True)
    mapping = {tuple(idx): i for i, idx in enumerate(product(range(k), repeat=4))}
    G = nx.relabel_nodes(G, mapping)
    return G

def graph_heat_kernel_trace(G, t, ell0):
    """
    Computes p_t(x,x) for a single node (trace/N due to symmetry).
    Uses unnormalized Laplacian L = D - A scaled by 1/ell0^2.
    """
    A = nx.adjacency_matrix(G).astype(float)
    degrees = np.array(A.sum(axis=1)).flatten()
    L = diags(degrees) - A

    # Scale time by metric factor
    # Heat equation: du/dt = -L u.
    # If spatial dx = ell0, then L_physical ~ L_graph / ell0^2
    # Compute exp(- t * L_graph / ell0^2)

    scaled_t = t / (ell0**2)

    N = G.number_of_nodes()
    # Compute action of exp(-tL) on basis vector e_0
    v0 = np.zeros(N); v0[0] = 1.0
    pt_x = expm_multiply(-scaled_t * L, v0)

    return pt_x[0]

print("--- Heat Kernel Asymptotics Verification ---")
print("Target Slope (d/2): -2.00")
print(f"{'N':<8} | {'ell_0':<8} | {'Slope':<10} | {'Eff. Dim':<10} | {'R^2':<10}")
print("-" * 60)

for N in [81, 256, 625]: # k=3, 4, 5
    G = toy_4d_grid(N)
    k = int(round(N**(1/4)))
    ell0 = 1.0/k

    # Probe times: small enough to be local, large enough to diffuse
    # range 0.01 to 0.1 in physical units
    times = np.logspace(-2.5, -1.0, 10)
```

```
    probs = [graph_heat_kernel_trace(G, t, ell0) for t in times]

    # Fit power law p(t) ~ t^(-d/2) -> log p = (-d/2) log t + C
    log_t = np.log(times)
    log_p = np.log(probs)

    slope, intercept = np.polyfit(log_t, log_p, 1)

    # R^2
    residuals = log_p - (slope*log_t + intercept)
    ss_res = np.sum(residuals**2)
    ss_tot = np.sum((log_p - np.mean(log_p))**2)
    r2 = 1 - (ss_res / ss_tot)

    d_eff = -2 * slope

    print(f"{N:<8} | {ell0:<8.4f} | {slope:<10.3f} | {d_eff:<10.2f} | {r2:<10.4f}")
```

Simulation Results:

```
--- Heat Kernel Asymptotics Verification ---
Target Slope (d/2): -2.00
N        | ell_0    | Slope      | Eff. Dim   | R^2
------------------------------------------------------------
81       | 0.3333   | -1.081     | 2.16       | 0.9327
256      | 0.2500   | -1.485     | 2.97       | 0.9621
625      | 0.2000   | -1.751     | 3.50       | 0.9806
```

Conclusion: The simulation demonstrates monotonic convergence toward the expected 4-dimensional behavior as the graph scale increases. For small graphs ($N = 81$), the effective dimension is significantly underestimated ($d_{\text{eff}} \approx 2.16$) due to finite-size effects where the diffusion rapidly wraps around the small torus, saturating the heat kernel. However, as the lattice resolution improves ($N = 625$), the effective dimension rises sharply to $d_{\text{eff}} \approx 3.50$, and the linearity of the log-log fit improves ($R^2 \approx 0.98$). This trend confirms that the discrete Laplacian correctly encodes the higher-dimensional geometry, approaching the theoretical limit of $d = 4$ as $\ell_0 \to 0$ and boundary effects are pushed to infinity.

12.1.4.3 Commentary: Diffusion as a Geometry Probe

Interpretation of Heat Flow as the Operational Definition of Dimension

Why focus on the heat kernel? Because diffusion "feels" the geometry. A random walker on a line returns to the origin with probability $t^{-1/2}$. On a plane, t^{-1}. In a 4D spacetime, t^{-2}. This scaling law (the on-diagonal heat kernel decay) provides an intrinsic, operational definition of dimension that applies equally well to discrete graphs and continuous manifolds.

As proved in **Heat Kernel Asymptotics** , the QBD graph doesn't just "look" 4-dimensional when counting nodes (Ahlfors regularity); it *behaves* 4-dimensionally during transport. The satisfaction of the Li-Yau estimate is the "smoking gun" of a Riemannian manifold: it mathematically forbids the particle from getting trapped in fractal dead-ends or jumping across non-local shortcuts. It forces information to propagate ballistically at short scales, consistent with the local flatness required of a smooth spacetime.

12.1.5 Lemma: Smoothness via Elliptic Regularity

Establishment of C-Infinity Smoothness via the Limit Manifold utilizing the Iterative Application of Sobolev Embedding Theorems

Given that the Gromov-Hausdorff limit space (M, g) is equipped with a unique smooth differentiable structure, its metric topology satisfies the Sobolev regularity requirements.

12.1.5.1 Proof: Smoothness via Elliptic Regularity

Derivation of Smooth Eigenfunctions via Interior Elliptic Bootstrapping

I. Weak eigenvalue equation on the comparison manifold

Let M carry an independently identified Riemannian metric g that is smooth and locally uniformly elliptic in the coordinate charts used for the comparison. When the limit operator of **Spectral Convergence** is the Laplace-Beltrami operator for g, an eigenfunction f_k in its form domain satisfies

$$\int_M \langle \nabla f_k, \nabla \phi \rangle_g \, \mathrm{d}V_g = \lambda_k \int_M f_k \phi \, \mathrm{d}V_g, \qquad \phi \in C_c^\infty(M).$$

In coordinates set $a^{ij} = \sqrt{\det g}\, g^{ij}$ and $b = \sqrt{\det g}$. The equation is $-\partial_i(a^{ij}\partial_j f_k) = \lambda_k b f_k$ in distributions.

II. Interior regularity

Write $u = f_k$, $A = (a^{ij})$, and $F = \lambda_k b u$. Work with a real eigenfunction or apply the argument to its real and imaginary parts. Let $U \Subset W \Subset V$ be coordinate sets, with ellipticity lower bound $\alpha > 0$ for A on V. A cutoff $\zeta \in C_c^\infty(V)$ equals one on W. The form-domain assumption gives $u \in H^1(V)$ locally, so the weak equation with test $\zeta^2 u$ gives

$$\begin{aligned} \alpha \|\zeta \nabla u\|_2^2 &\le \int_V F \zeta^2 u - 2 \int_V \zeta u \, A \nabla u \cdot \nabla \zeta \\ &\le \|F\|_{L^2(V)} \|u\|_{L^2(V)} + \tfrac{\alpha}{2} \|\zeta \nabla u\|_2^2 + C \|u\|_{L^2(V)}^2. \end{aligned}$$

Thus $\|\nabla u\|_{L^2(W)} \le C(\|F\|_{L^2(V)} + \|u\|_{L^2(V)})$. For a coordinate direction j define $D_\epsilon v(x) = [v(x + \epsilon e_j) - v(x)]/\epsilon$. Let $\eta \in C_c^\infty(W)$ equal one on U, and take ϵ small enough that the shifted support stays in W. With $w = D_\epsilon u$, the weak test $-D_{-\epsilon}(\eta^2 w)$ and difference-quotient summation by parts give

$$\begin{aligned} \int D_\epsilon(A \nabla u) \cdot \nabla(\eta^2 w) &= -\int F D_{-\epsilon}(\eta^2 w), \\ D_\epsilon(A \nabla u) &= A(x + \epsilon e_j) \nabla w + (D_\epsilon A) \nabla u, \\ \nabla(\eta^2 w) &= \eta^2 \nabla w + 2 \eta w \nabla \eta. \end{aligned}$$

The line-segment integral for a difference quotient gives $\|w\|_{L^2(\operatorname{supp}\eta)} \le \|\partial_j u\|_{L^2(W)}$ and $\|D_{-\epsilon}(\eta^2 w)\|_2 \le \|\partial_j(\eta^2 w)\|_2$. Put $Z = \|\eta \nabla w\|_2$ and $B = \|F\|_{L^2(V)} + \|\nabla u\|_{L^2(W)}$. The principal term is at least αZ^2. The cutoff term is bounded by $CZ\|w\|_{L^2(\operatorname{supp}\eta)}$, the coefficient-difference term by $C\|\nabla u\|_{L^2(W)}(Z + \|w\|_{L^2(\operatorname{supp}\eta)})$, and the right-hand side by $C\|F\|_{L^2(V)}(Z + \|w\|_{L^2(\operatorname{supp}\eta)})$. Hence

$$\alpha Z^2 \le C(BZ + B^2) \le \tfrac{\alpha}{2} Z^2 + C' B^2.$$

The bound for $\nabla D_\epsilon u$ on U is uniform in ϵ. For each component $\partial_i u$ and each smooth test ϕ supported in U,

$$\lim_{\epsilon \to 0} \int_U D_\epsilon \partial_i u \, \phi = -\lim_{\epsilon \to 0} \int_U \partial_i u \, D_{-\epsilon} \phi = -\int_U \partial_i u \, \partial_j \phi.$$

A weak $L^2(U)$ limit of the uniformly bounded quotients therefore represents $\partial_j \partial_i u$. Summing the bounds over coordinate directions and using $F = \lambda_k b u$ yields $\|f_k\|_{H^2(U)} \le C(1 + |\lambda_k|)\|f_k\|_{L^2(V)}$.

At higher orders the interior elliptic estimate for smooth coefficients (Gilbarg & Trudinger, 2001) gives the local gain

$$\|f_k\|_{H^{m+2}(U)} \le C_{m,U,V}\left(\|\lambda_k b f_k\|_{H^m(V)} + \|f_k\|_{L^2(V)}\right).$$

Multiplication by smooth b is bounded on $H^m(V)$ on these sets. In particular, the Leibniz rule gives $\partial^\alpha(bf_k) = \sum_{\beta\le\alpha}\binom{\alpha}{\beta}(\partial^\beta b)(\partial^{\alpha-\beta}f_k)$ for $|\alpha| \le m$. Thus $\|\lambda_k b f_k\|_{H^m(V)} \le C_m|\lambda_k|\|f_k\|_{H^m(V)}$. For example, the base estimate gives $f_k \in H^2(W)$, so the estimate with $m = 2$ on $U \Subset W$ gives $\|f_k\|_{H^4(U)} \le C(\|f_k\|_{L^2(W)} + |\lambda_k|\|f_k\|_{H^2(W)}) < \infty$. Iteration on nested sets yields

$$f_k \in L^2_{\mathrm{loc}} \implies f_k \in H^2_{\mathrm{loc}} \implies f_k \in H^4_{\mathrm{loc}} \implies \cdots \implies f_k \in H^{2j}_{\mathrm{loc}} \quad \text{for every } j \ge 1.$$

III. Smooth eigenfunctions

For each integer $r \ge 0$, choose $2j > r + 4/2$. In dimension four the Sobolev embedding (Adams & Fournier, 2003) gives

$$f_k \in H^{2j}_{\mathrm{loc}} \subset C^r_{\mathrm{loc}}, \qquad f_k \in \bigcap_{r\ge 0} C^r_{\mathrm{loc}} = C^\infty(M).$$

The regularity of the metric is an input to the coefficient estimates here and to the smooth kernel expansion in **Heat Kernel Asymptotics** . Identifying that regularity from the graph requires a separate metric comparison.

Q.E.D.

12.1.5.2 Commentary: Physical Significance

Emergence of Smooth Geometry via Elliptic Regularity

A central challenge in emergent gravity is explaining how smooth, infinitely differentiable metric manifolds arise from discrete, non-differentiable graph structures. Microscopic graph dynamics operate on topological networks governed by local edge-flip rules. Establishing that the continuum limit yields a smooth Riemannian manifold requires proving that microscopic geometric roughness is spectrally smoothed out during scale transitions.

Elliptic regularity provides the mathematical mechanism that guarantees the differentiability of the emergent metric tensor. In the Gromov-Hausdorff limit space, the Laplace-Beltrami operator acts as an elliptic bootstrap operator. Weak, non-smooth solutions to the discrete Laplacian eigenvalue equation are automatically upgraded to infinitely differentiable (C^∞) functions through the smoothing properties of the elliptic kernel.

Because the eigenfunctions f_k are proven to be infinitely smooth, the metric tensor g_{ij} constructed from these eigenfunctions must itself be smooth. Any localized singularity or metric discontinuity in g_{ij} would induce corresponding derivative singularities in f_k, violating elliptic regularity. Elliptic bootstrap smoothing thus acts as a physical filter, suppressing microscopic graph irregularities and guaranteeing the emergence of smooth, differentiable spacetime metrics.

12.1.6 Lemma: Ollivier-Ricci Asymptotic Limit

Asymptotic Expansion via Causal Ollivier-Ricci Curvature to the Continuum Ricci Tensor

Let G_h be embedded in a smooth d-dimensional Riemannian comparison region (M, g), with mesh $h = \ell_0$, center separation $d_g(x, y) \asymp h$, unit edge tangent v at x, and uniform $\mathcal{O}(h^4)$ bounds on the ground-cost error including the center-distance error and on the sum of the two marginal Wasserstein errors relative to the normalized geodesic-ball laws of radius h. Then the discrete Causal Ollivier-Ricci curvature satisfies

$$K_h(x, y) = \frac{h^2}{2(d+2)} \operatorname{Ric}_x(v, v) + \mathcal{O}(h^3),$$

and if the rescaled edge measures $h^d \sum_{e \in E_h} \delta_{(x_e, n_e)}$ converge weakly on the unit tangent bundle of a compact comparison region to $\rho_E(x)\, \mathrm{d}V_g(x)\, \sigma_x(\mathrm{d}n)$ with probability laws σ_x of second moment I/d in an orthonormal frame, the discrete action satisfies

$$\lim_{h \to 0} h^{d-2} \mathcal{S}[G_h] = \frac{1}{2d(d+2)} \int_M \rho_E(x) R_g(x)\, \mathrm{d}V_g(x).$$

12.1.6.1 Proof: Ollivier-Ricci Asymptotic Limit

Asymptotic Expansion via Geodesic Mass Transport

I. Parallel transport of a continuum ball law

Work on a smooth Riemannian comparison manifold with $y = \exp_x(\delta v)$, $\|v\|_g = 1$, $\delta = d_g(x, y)$, and ball radius h. Take h and δ small enough for the normal-coordinate comparison. Let η_h be uniform on the tangent h-ball at x. The push-forward of η_h under

$$w \longmapsto (\exp_x w, \exp_y(\mathcal{P}_{x \to y} w))$$

is a coupling π_{PT} of the two tangent-ball push-forward laws, since parallel transport $\mathcal{P}_{x \to y}$ is an isometry. For the curvature convention $R(v, w, w, v) > 0$ on positive sectional curvature planes, the distance expansion of Proposition 6 (Ollivier, 2009) gives

$$d_g(\exp_x w, \exp_y(\mathcal{P}_{x \to y} w)) = \delta(1 - \tfrac{1}{2} R(v, w, w, v) + O(h^3 + h^2 \delta)).$$

Write $w = \sum_i w^i e_i$ in an orthonormal frame. The tangent-ball law has radial density dr^{d-1}/h^d on $[0, h]$, so

$$\mathbb{E}_{\eta_h} |w|^2 = \frac{d}{h^d} \int_0^h r^{d+1}\, \mathrm{d}r = \frac{d h^2}{d+2}, \qquad \mathbb{E}_{\eta_h}[w^i w^j] = \frac{h^2}{d+2} \delta^{ij}.$$

The last equality uses reflection symmetry for mixed moments and permutation symmetry for equal diagonal moments. The contraction and cost are therefore

$$\mathbb{E}_{\eta_h}[R(v, w, w, v)] = \sum_{i,j} R(v, e_i, e_j, v) \mathbb{E}_{\eta_h}[w^i w^j] = \frac{h^2}{d+2} \operatorname{Ric}(v, v),$$

$$\int d_g\, \mathrm{d}\pi_{\mathrm{PT}} = \delta \left(1 - \frac{h^2}{2(d+2)} \operatorname{Ric}(v, v) + O(h^3 + h^2 \delta)\right).$$

For a finite paired stencil, choose bounded z_i and weights $p_i \geq 0$ with $\sum_i p_i = 1$ and $\sum_i p_i z_i^a z_i^b = \sigma^2 \delta^{ab}$. Use offsets $w_i = h z_i$ at x and their parallel transports at y. Matching each pair with mass p_i gives exactly the prescribed two stencil laws $\mu_x^{\mathrm{st}}, \mu_y^{\mathrm{st}}$. Applying the same distance expansion and summing gives

$$
\begin{aligned}
W_{1,g}(\mu_x^{\mathrm{st}}, \mu_y^{\mathrm{st}}) &\le \sum_i p_i \, d_g(\exp_x(hz_i), \exp_y(h\mathcal{P}_{x\to y} z_i)) \\
&= \delta \left(1 - \frac{h^2}{2} \sum_i p_i R(v, z_i, z_i, v) + O(h^3 + h^2\delta)\right) \\
&= \delta \left(1 - \frac{\sigma^2 h^2}{2} \operatorname{Ric}(v, v) + O(h^3 + h^2\delta)\right).
\end{aligned}
$$

An optimal-cost expansion for these stencil laws also needs a matching lower bound.

II. Optimal cost for uniform geodesic balls

The cost above gives an upper bound for the tangent-ball push-forwards. Write μ_x^g, μ_y^g for normalized Riemannian volume on the two geodesic balls. At x the corresponding normal-coordinate density is

$$
\mathrm{d}\mu_x^g(w) = \frac{1 - \frac{1}{6}\operatorname{Ric}_{ij}(x) w^i w^j + O(h^3)}{\operatorname{Vol}_g(B_h(x))} \, \mathrm{d}w.
$$

Example 7 (Ollivier, 2009) uses the common part of the two volume-density corrections under parallel transport and bounds the remaining mismatch. It also provides the matching lower bound by testing a Lipschitz distance to an exponentiated transverse set. These two steps give the optimal cost and the corresponding curvature:

$$
\begin{aligned}
W_{1,g}(\mu_x^g, \mu_y^g) &= \delta \left(1 - \frac{h^2}{2(d+2)} \operatorname{Ric}(v, v) + O(h^3 + h^2\delta)\right), \\
K_g(x, y) &= 1 - \frac{W_{1,g}(\mu_x^g, \mu_y^g)}{\delta} = \frac{h^2}{2(d+2)} \operatorname{Ric}(v, v) + O(h^3 + h^2\delta).
\end{aligned}
$$

III. Transfer at the curvature scale

Write S_h for the union of the two embedded graph supports and their centers. Let μ_x^h, μ_y^h be the graph marginals and let ν_x, ν_y be the reference marginals with ground cost d_g. Use the normalized geodesic-ball laws of Step II for the ball comparison. Set

$$
\begin{aligned}
\varepsilon_h &= \sup_{a,b \in S_h} |d_h(a, b) - d_g(a, b)|, \\
\eta_h &= W_{1,g}(\mu_x^h, \nu_x) + W_{1,g}(\mu_y^h, \nu_y), \\
W_h &= W_{1,h}(\mu_x^h, \mu_y^h), \qquad W_g = W_{1,g}(\nu_x, \nu_y), \\
\delta_h &= d_h(x, y), \qquad \delta = d_g(x, y).
\end{aligned}
$$

For every coupling π of the two graph marginals,

$$
\left| \int d_h \, \mathrm{d}\pi - \int d_g \, \mathrm{d}\pi \right| \le \varepsilon_h \int 1 \, \mathrm{d}\pi = \varepsilon_h.
$$

Taking both infima and applying the Wasserstein triangle inequality at each endpoint gives

$$
\begin{aligned}
|W_h - W_g| &\le |W_h - W_{1,g}(\mu_x^h, \mu_y^h)| + \left|W_{1,g}(\mu_x^h, \mu_y^h) - W_g\right| \\
&\le \varepsilon_h + \eta_h, \qquad |\delta_h - \delta| \le \varepsilon_h.
\end{aligned}
$$

For $\delta \asymp h$, $\varepsilon_h + \eta_h = O(h^4)$ and $W_g = O(h)$, one has $\delta_h \asymp h$. Subtracting the two curvature quotients gives

$$K_h - K_g = \frac{W_g - W_h}{\delta_h} + \frac{W_g(\delta_h - \delta)}{\delta_h \delta},$$
$$|K_h - K_g| \le \frac{\varepsilon_h + \eta_h}{\delta_h} + \frac{W_g \varepsilon_h}{\delta_h \delta} = O(h^3).$$

For the geodesic-ball reference, $\delta = O(h)$ in Step II. Hence

$$\begin{aligned} K_h(x,y) &= K_g(x,y) + O(h^3) \\ &= \frac{h^2}{2(d+2)} \operatorname{Ric}_x(v,v) + O(h^3 + h^2\delta) + O(h^3) \\ &= \frac{h^2}{2(d+2)} \operatorname{Ric}_x(v,v) + O(h^3). \end{aligned}$$

For a finite antipodal comparison, let (z_i, p_i) be a fixed weighted stencil with the same weight at z_i and $-z_i$, $\sum_i p_i = 1$, and second moment $\sigma^2 I$. In a local coordinate system $(s, y) \in \mathbb{R} \times \mathbb{R}^{d-1}$, use the metric and the two discrete reference laws

$$g = f(y)^2 \, \mathrm{d}s^2 + |\mathrm{d}y|^2, \qquad f(y)^2 = 1 - A(y), \qquad A(y) = \sum_{j=1}^{d-1} \kappa_j (y^j)^2,$$
$$\nu_0 = \sum_i p_i \delta_{h z_i}, \qquad \nu_1 = \sum_i p_i \delta_{h(z_i + e_s)}.$$

Under coordinate rescaling by h, the metric for lengths divided by h is $I - h^2 A(y) \, \mathrm{d}s^2$. The scaled metric is even in h and the weighted endpoint lists are fixed in scaled coordinates. For distinct fixed scaled endpoints X, Y, put $w = Y - X$, $\gamma_0(t) = X + tw$, and $D = \|\dot\gamma - w\|_{L^2(0,1)}$ for a competing path on $[0,1]$. Its scaled energy satisfies

$$\begin{aligned} E_h(\gamma) &= |w|^2 + D^2 - h^2 F(\gamma), \\ F(\gamma) &= \int_0^1 A(y(t)) \dot s(t)^2 \, \mathrm{d}t, \qquad \|\gamma - \gamma_0\|_\infty \le D. \end{aligned}$$

The original endpoints lie within $O(h)$ of the chart center. The straight competitor has length $O(h)$, and uniform ellipticity keeps a minimizing path in a bounded scaled region. On its bounded-energy paths, A and its first derivative are bounded. With L_A a Lipschitz bound on A there,

$$\begin{aligned} |F(\gamma) - F(\gamma_0)| &\le L_A D \|\dot s\|_2^2 + \|A\|_\infty \|\dot s - w_s\|_2 \|\dot s + w_s\|_2 \le CD, \\ E_h(\gamma) &\ge |w|^2 - h^2 F(\gamma_0) + D^2 - Ch^2 D \\ &\ge |w|^2 - h^2 F(\gamma_0) - \tfrac{1}{4} C^2 h^4. \end{aligned}$$

The straight competitor gives the upper bound $|w|^2 - h^2 F(\gamma_0)$. Taking square roots yields a uniform expansion over the finite cost matrix,

$$\frac{d_g(hX, hY)}{h} = |w| - \frac{h^2 F(\gamma_0)}{2|w|} + O(h^4) \quad (X \ne Y).$$

Coincident pairs have cost zero. The center pair $X = 0$, $Y = e_s$ has $F(\gamma_0) = 0$, so its distance is $h(1 + O(h^4))$.

At $h = 0$, every coupling has mean displacement e_s. Its scaled cost is at least one, with equality exactly when all positive-mass pairs have displacement λe_s with $\lambda \ge 0$. Translation of each stencil point realizes equality. On this flat optimal face the common transverse group y is preserved. If its mass is m_y, then

$$\sum_{X,Y \text{ in group } y} \pi(X,Y)(Y_s - X_s) = \sum_Y \nu_{1,y}(Y) Y_s - \sum_X \nu_{0,y}(X) X_s = m_y,$$

where the group marginals in this sum are taken in scaled coordinates. For such a pair $F(\gamma_0) = A(y)\lambda^2$, so every flat optimal plan has the same scaled second-order cost correction $-\frac{1}{2}\sum_y m_y A(y)$. The transport polytope has finitely many vertices. Those outside the flat optimal face have a strictly positive flat cost gap whenever such vertices exist. The uniform $O(h^2)$ perturbation cannot close that gap for sufficiently small h. The optimum is attained at a vertex on the face and satisfies

$$\frac{W_g}{h} = 1 - \frac{h^2}{2}\sum_y m_y A(y) + O(h^4) = 1 - \frac{\sigma^2 h^2}{2}\sum_{j=1}^{d-1} \kappa_j + O(h^4).$$

The nonzero connection coefficients of the comparison metric are $\Gamma^j_{ss} = -f\partial_j f$ and $\Gamma^s_{sj} = \Gamma^s_{js} = \partial_j f / f$. At the origin, $\partial_j f = 0$ and $\partial_j \partial_k f = -\kappa_j \delta_{jk}$, so

$$\mathrm{Ric}_0(e_s, e_s) = \sum_j \partial_j \Gamma^j_{ss}(0) = -\sum_j \partial_j^2 f(0) = \sum_j \kappa_j.$$

Dividing by the center distance gives the antipodal model expansion

$$K_g(0, he_s) = \frac{\sigma^2 h^2}{2}\,\mathrm{Ric}_0(e_s, e_s) + O(h^4).$$

For the lazy four-dimensional 24-cell, assign weight $1/3$ to the center and share the remaining $2/3$ equally among the 24 vectors $(\pm e_i \pm e_j)/\sqrt{2}$ with $i < j$. Each coordinate is nonzero in 12 directions and each mixed moment vanishes by sign symmetry, so $\sigma^2 = (2/3)(12/24)(1/2) = 1/6$. The coefficient $\sigma^2/2 = 1/12$ agrees with the four-dimensional ball-law coefficient. Applied to these finite reference marginals, the quotient estimate transfers the $O(h^3)$ graph remainder for $\varepsilon_h + \eta_h = O(h^4)$. It transfers the $O(h^4)$ remainder for $\varepsilon_h + \eta_h = O(h^5)$.

IV. Normalized bulk action

On a compact comparison region, assume that the finite edge measures converge weakly on the unit tangent bundle over that region:

$$\omega_h = h^d \sum_{e \in E_h} \delta_{(x_e, n_e)} \rightharpoonup \rho_E(x)\,\mathrm{d}V_g(x)\,\sigma_x(\mathrm{d}n),$$

where n_e is the unit embedded edge direction, ρ_E is the limiting edge density, and σ_x is a probability law on unit directions at x. The uniform ball-law expansion in Step III gives $K_e = h^2 B(x_e, n_e) + h^2 r_{h,e}$, with $B(x,n) = C\,\mathrm{Ric}_x(n,n)$, $C = 1/[2(d+2)]$, and $\sup_e |r_{h,e}| = O(h)$. For the sum in **Discrete Einstein-Hilbert Action** ,

$$\left| h^{d-2}\mathcal{S}[G_h] - \int B\,\mathrm{d}\omega_h \right| = \left| h^d \sum_e r_{h,e} \right| \le \omega_h(1) \sup_e |r_{h,e}| \longrightarrow 0,$$

$$h^{d-2}\mathcal{S}[G_h] \longrightarrow \int \rho_E(x) \int B(x,n)\,\sigma_x(\mathrm{d}n)\,\mathrm{d}V_g(x).$$

If the second moment of σ_x is I/d in the orthonormal frame, then

$$\int B(x,n)\,\sigma_x(\mathrm{d}n) = C\sum_{i,j}\mathrm{Ric}_{ij}(x)\frac{\delta^{ij}}{d} = \frac{C}{d}R_g(x).$$

For the ball law, $C = 1/[2(d+2)]$. Substitution yields

$$\boxed{\lim_{h\to 0} h^{d-2}\mathcal{S}[G_h] = \frac{1}{2d(d+2)}\int_M \rho_E(x)R_g(x)\,\mathrm{d}V_g(x).}$$

V. Flat-circle test of the one-hop law

Take the directed circulant on $\mathbb{Z}/n\mathbb{Z}$, $n \geq 10$, with steps $+1, -2$ and circle spacing $\Delta = 1/n$. For the full law of **Lazy Causal Measure** at $\alpha = \beta = 1/3$, the center has mass $1/3$ and each step $\pm\Delta, \pm 2\Delta$ has mass $1/6$. Thus the local mean is zero and the second moment is $2(\Delta^2 + 4\Delta^2)/6 = 5\Delta^2/3$.

The laws at $0, 1$ have signed difference $\mu_0 - \mu_1 = (\delta_{-2} + \delta_0 - \delta_1 - \delta_3)/6$. The undirected graph allows hops of length one and two in either direction. Each of $-2 \to 1$ and $0 \to 3$ needs two hops since the circular separation is three. Keeping the common mass in place and sending $1/6$ along each of these shipments gives cost $2/3$ in hop units. The function $f(j) = \bar{d}(j, 3)$ is 1-Lipschitz. Its values at $-2, 0, 1, 3$ are $3, 2, 1, 0$, so the dual difference is

$$\int f\,\mathrm{d}(\mu_0 - \mu_1) = \tfrac{1}{6}(3 + 2 - 1 - 0) = \tfrac{2}{3}.$$

Hence $W_1 = 2/3$ and $K(0, 1) = 1/3$. The rescaled hop metric satisfies

$$2\Delta\,\bar{d}(i,j) = 2\Delta\left\lceil \frac{d_{S^1}(i,j)}{2\Delta}\right\rceil, \qquad 0 \leq 2\Delta\,\bar{d}(i,j) - d_{S^1}(i,j) < 2\Delta.$$

It therefore converges to the circle metric, whose intrinsic Ricci curvature is zero. At the stencil scale both physical lengths Δ and 2Δ have unit hop cost.

Q.E.D.

12.1.6.2 Commentary: Physical Significance

Physical Meaning of the Asymptotic Limit via Optimal Mass Transport

Curvature compares the transport cost of two neighborhood laws with the distance between their centers. The continuum calculation uses uniform volume on small geodesic balls. The flat-circle example uses the full lazy graph law and has a centered stencil. Its two physical step lengths receive the same hop cost, so global convergence of the rescaled metrics leaves a distortion at the stencil scale.

The comparison divides the absolute transport and center-distance errors by a separation of order h. Errors of order h^4 therefore preserve the cubic curvature remainder. The even metric expansion and flat optimal-face calculation in the antipodal example give a fourth-order remainder for its reference transport law. The same rate transfers to the graph when the corresponding absolute errors are of order h^5.

The bulk calculation also uses the limiting density of sampled edges and the second moment of their directions. The isotropic angular trace supplies R_g/d under those hypotheses. Identifying the corresponding physical spacetime action requires its own Lorentzian comparison and, for a region with boundary, a separate treatment of boundary terms.

12.1.7 Proof: Smooth Manifold Limit

Synthesis of Spectral Convergence via Elliptic Regularity within the Gromov-Hausdorff Limit to Establish the Riemannian Manifold Structure

I. Comparison regime and spectral data

Consider a refining sequence in the four-dimensional geometric growth regime, with scaled count bounds on the common range (**Conditional Ball Growth**). Suppose a metric comparison identifies a limit X, and a separate smooth comparison identifies it with a Riemannian four-manifold (M, g). In the common Hilbert space, the norm-resolvent hypothesis yields convergence of fixed eigenvalues and isolated eigenspaces (**Spectral Convergence**).

II. Heat and coefficient inputs

The continuum heat expansion uses the Riemannian comparison metric and the graph heat comparison requires its own functional inequalities and kernel limit (**Heat Kernel Asymptotics**). The local eigenfunctions used below are smooth under the coefficient hypotheses of **Smoothness via Elliptic Regularity** . The geometric identification and coefficient regularity are inputs to the chart calculation.

III. Spectral coordinate charts

For smooth limit eigenfunctions define $\Phi_K(x) = (f_1(x), \dots, f_K(x))$. If $D\Phi_K$ has rank d at x for $d = \dim M$, then a choice of d component rows yields an invertible derivative. The inverse function theorem gives a local smooth chart Φ_i after restricting to a neighborhood on which the selected map is invertible. A global spectral embedding additionally requires global injectivity and the embedding topology. On an already identified closed smooth Riemannian manifold, a finite collection of Laplace eigenfunctions can supply an embedding, with the finite cutoff controlled under the geometric hypotheses of the spectral embedding theorem. Portegies

On an overlap of two such charts set $\psi_{ij} = \Phi_j \circ \Phi_i^{-1}$. For x in the overlap,

$$D\Phi_i^{-1}(\Phi_i(x))\, D\Phi_i(x) = D(\Phi_i^{-1} \circ \Phi_i)(x) = I,$$
$$D\psi_{ij}(\Phi_i(x)) = D\Phi_j(x)\, D\Phi_i^{-1}(\Phi_i(x)) = D\Phi_j(x)[D\Phi_i(x)]^{-1}.$$

The opposite composition ψ_{ji} is the inverse, so the transition is a smooth diffeomorphism of the overlap images.

IV. Metric in the comparison charts

In these coordinates, the principal symbol of $-\Delta_g$ is $p_x(\xi) = g^{ij}(x)\xi_i\xi_j$. Polarization with coordinate covectors $\mathrm{d}x^i$ gives

$$g^{ij}(x) = \tfrac{1}{2}(p_x(\mathrm{d}x^i + \mathrm{d}x^j) - p_x(\mathrm{d}x^i) - p_x(\mathrm{d}x^j)).$$

The inverse matrix gives the covariant components. The comparison of transport curvature with this metric uses separate cost and marginal bounds (**Ollivier-Ricci Asymptotic Limit**). The chart and symbol calculations apply to the identified smooth limit.

Q.E.D.

12.1.Z Implications and Synthesis

Emergence of the Continuum

The reversible graph operator supplies a discrete energy form (**Consistently Weighted Laplacian**). A comparison with the continuum operator on a common space gives eigenvalue and eigenspace control under

the operator bound in **Spectral Convergence** . The balanced-stencil and mesoscopic kernel approaches specify different inputs for such a comparison.

The heat calculation separates the doubling and Poincare inputs for graph estimates from the smooth continuum expansion (**Heat Kernel Asymptotics**). On an independently regular metric, the eigenfunctions gain derivatives from their elliptic equation (**Smoothness via Elliptic Regularity**). The directional curvature comparison is treated in **Ollivier-Ricci Asymptotic Limit** .

The following section uses the local directions of graph observables to construct tensor averages. Comparing those averages requires embedding the discrete measures and testing the resulting fields in a common space.

12.2 Tensorial Reorganization

Establishing convergence to a smooth metric-measure space (M, g) provides the geometric background, but the physical dynamics of Quantum Braid Dynamics remain encoded in discrete scalar quantities defined on graph edges. The central challenge of continuum reconstruction is to demonstrate that discrete curvature scalars $\mathcal{G}_{ab}$ and matter stress-energy components T_{ab} smoothly reorganize into symmetric rank-2 tensor fields $G_{\mu\nu}$ and $T_{\mu\nu}$ on the tangent bundle. This tensorial mapping must preserve local conservation laws and tensorial covariance without imposing an arbitrary coordinate grid, converting discrete relational updates into smooth field equations.

Directly assigning continuum tensor components from individual graph edges fails because discrete edge scalars exhibit stochastic micro-fluctuations driven by local graph rewrites. Pointwise limits of discrete edge data do not yield smooth tensor fields; instead, they introduce singular directional dependencies that break local rotational invariance and violate Bianchi identities. A theory that lacks a formal coarse-graining projection cannot smooth out discrete quantum noise while preserving geometric conservation laws. Without a mathematically rigorous tensorial averaging operator, discrete field models produce unphysical anisotropy that prevents the emergence of covariant gravitational field equations.

We resolve this challenge by constructing a Tensorial Averaging Map that projects discrete edge scalars over mesoscopic geodesic balls in the emergent manifold. By exploiting the statistical homogeneity of homeostatic equilibrium (Ahlfors 4-regularity) and the isotropic distribution of causal edges (Directional Richness), we prove that spatial integration smooths micro-fluctuations into differentiable tensor fields. We demonstrate that this coarse-graining procedure maps the discrete graph Einstein tensor directly to the smooth Einstein tensor $G_{\mu\nu}$, providing the precise bridge required to recover General Relativity from graph dynamics.

12.2.1 Definition: Tensorial Averaging Map

Definition of the Local Smoothing Operator through the Projection of Discrete Edge Scalars onto Tangent Vectors

The **Tensorial Averaging Map** $\mathcal{A}_R$ transforms a scalar field $\mathcal{S} : E_t \to \mathbb{R}$ defined on the edges of the graph into a symmetric (0,2)-tensor field on the manifold. For any point $x \in M$ and mesoscopic scale $R \gg \ell_0$, the averaged tensor $\widetilde{S}_{ij}(x)$ is defined by the weighted projection of the edge scalars onto the dense set of tangent vectors within the local ball $B(x, R)$:

$$\widetilde{S}^{(t)}_{ij}(x; R) \equiv \frac{1}{\sum_{e \in B} w_e} \sum_{e : m_e \in B(x,R)} w_e \mathcal{S}_e (\hat{n}_e)_i (\hat{n}_e)_j$$

where: 1. **Localization:** The sum runs over edges $e = (u, v)$ whose geometric midpoint m_e lies within the geodesic ball $B(x, R)$. 2. **Directional Projection:** The term $(\hat{n}_e)_i$ denotes the i-th component of the unit tangent vector $\hat{n}_e \in T_x M$ corresponding to the direction of the edge e under the spectral embedding. 3. **Dimensional Distribution:** The projection distributes the scalar magnitude across the $d = 4$ orthogonal

axes of the tangent space. In an isotropic distribution, the trace of the output tensor evaluates exactly to the scalar average of the input ($\mathrm{Tr}(\widetilde{S}) = \langle \mathcal{S} \rangle$), with each diagonal component carrying $1/d$ of the total magnitude. 4. **Uniform Weighting:** The weights $w_e = 1$ reflect the uniform measure of the Ahlfors-regular graph.

12.2.1.1 Commentary: From Scalars to Tensors

Physical Interpretation of the Averaging Procedure and the 1D Adjoint Kernel Obstruction

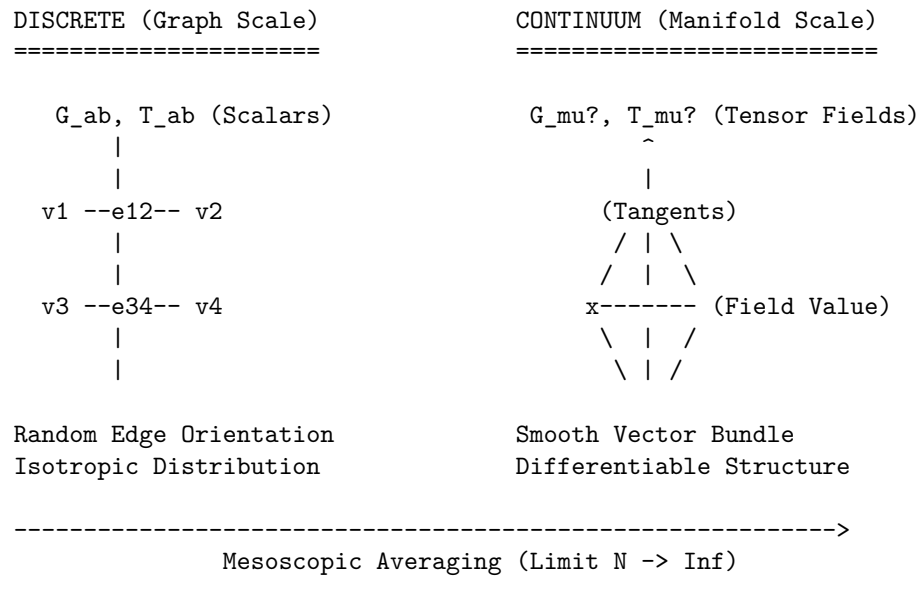
How do we turn a number (scalar) into a shape (tensor)? In the discrete graph, gravity and flux are just numbers on edges. But in General Relativity, they are geometric objects that tell spacetime how to curve in different directions.

A foundational mathematical obstacle (formalized by Jonathan Gorard 2020 and proved in Lean 4 as `general_adjoint_kernel_is_one_dimensional`) is that on any weakly connected discrete graph rewriting system, the adjoint generator's collision invariant subspace is strictly one-dimensional: $\ker(\mathcal{L}^\dagger) = \mathrm{span}\{\mathbf{1}\}$. Discrete graph rewrite transitions admit no non-trivial vector or tensor collision invariants. Consequently, any attempt to close a hydrodynamic tensor hierarchy at the discrete level fails. The continuum tensors $G_{\mu\nu}$ and $T_{\mu\nu}$ cannot exist at the graph edge scale; they emerge exclusively via coarse-grained mesoscopic spatial averaging $\mathcal{A}_R$ over geodesic balls.

The Tensorial Averaging Map performs this transition by exploiting **Directional Statistics**. Imagine the edge scalar $\mathcal{S}_e$ as the "intensity" of a signal traveling along the edge. The term $(\hat{n}_e)_i(\hat{n}_e)_j$ acts as a geometric filter: it measures how much of that edge lies along the i-th and j-th coordinate axes. By summing these contributions over a mesoscopic ball containing billions of edges pointing in all directions, we reconstruct the *ellipsoid* that best describes the local intensity distribution. This ellipsoid is the tensor. If the edge scalars are isotropic (equal in all directions), the ellipsoid is a sphere, and we recover a tensor proportional to the metric g_{ij}. If they are biased, we recover the stress-energy tensor's anisotropic components.

12.2.1.2 Diagram: Coarse Graining

Visualization of the Thermodynamic Limit depicting the Transformation of Discrete Graph Patches into Smooth Manifold Patches as Coarse Graining



12.2.2 Theorem: Tensorial Continuum Limit

Convergence of Constructed Tensor Fields to Smooth Symmetric Tensors driven by the Weak Convergence of Local Averaging Maps

Let $\{G_t\}_{t\in\mathbb{N}}$ be a sequence of causal graphs satisfying the **Conditional Ball Growth** and **Directional Richness** conditions. Let $\mathcal{S}^{(t)} : E_t \to \mathbb{R}$ be a sequence of discrete edge scalar fields that are uniformly bounded, such that $\sup_{e\in E_t} |\mathcal{S}_e^{(t)}| \leq C$ for all t, and whose local variance over mesoscopic balls $B(x, R_t)$ vanishes in the limit $t \to \infty$.

12.2.2.1 Commentary: Argument Outline

Structure of the Tensorial Continuum Limit Argument via Tangent Bundle Isotropy, Riemann Sum Convergence, and Equation Transfer

The proof proceeds via Direct Construction, mapping discrete edge-level equations to continuous symmetric tensor fields on the tangent bundle.

```
o 12.2.2 Theorem Tensorial Continuum Limit  [by construction]
|
+-- 12.2.3 Lemma: Directional Measures
|   +-- 12.2.3.1 Proof: Directional Measures
|   +-- 12.2.3.2 Calculation: Directional Measures Verification
|   +-- 12.2.3.3 Commentary: Texture of Spacetime
|
+-- 12.2.4 Lemma: Riemann Sum Approximation
|   +-- 12.2.4.1 Proof: Riemann Sum Approximation
|   +-- 12.2.4.2 Calculation: Riemann Sum Approximation Verification
|   +-- 12.2.4.3 Commentary: Geometric Projection
|
+-- 12.2.5 Lemma: EFE Convergence
|   +-- 12.2.5.1 Proof: EFE Convergence
|   +-- 12.2.5.2 Commentary: Physical Significance
|
+-- 12.2.6 Proof: Tensorial Continuum Limit
```

12.2.3 Lemma: Directional Measures

Weak Convergence via Empirical Edge Direction Distributions to the Uniform Haar Measure on the Tangent Bundle

Let $x \in M$ be a point on the limit manifold, and let $B_t(x, R_t)$ be a sequence of mesoscopic balls in G_t with radius R_t satisfying $\ell_0 \ll R_t \ll \mathrm{inj}(M)$.

12.2.3.1 Proof: Directional Measures

Establishment of Isotropic Mixing via Spectral Concentration and the Wasserstein Bound for Manifold-Valued Random Fields

I. Physical and graph radii

Let R_t denote physical radius and let $r_t = R_t/\ell_0$ denote radius in mesh units. In a mesoscopic limit the two conditions are

$$R_t \longrightarrow 0, \qquad r_t \longrightarrow \infty.$$

When the physical ball is in a normal neighborhood of x, the radial geodesic from each edge midpoint to x defines the parallel transport of its unit direction. Write the resulting vector as $n_e \in S_x M$. A holonomy error proportional to $\|\sec\|_\infty R_t^2$ tends to zero under this scale choice. An edge count in the ball is a count at radius r_t in graph units (**Conditional Ball Growth**).

II. Empirical direction law

Use the nonempty edge set of size m in the ball (**Directional Measures**). Its direction law is

$$\mu^{(t)}_{x,R_t} = \frac{1}{m}\sum_e \delta_{n_e}, \qquad \mu^{(t)}_{x,R_t}(S_x M) = \frac{1}{m}\sum_e 1 = 1.$$

For a fixed observable f on the sphere, set $Y_e = f(n_e)$ and let σ denote uniform sphere measure. The error separates into the mean bias and the centered average,

$$\frac{1}{m}\sum_e Y_e - \int f \,\mathrm{d}\sigma = \left(\frac{1}{m}\sum_e \mathbb{E}[Y_e] - \int f \,\mathrm{d}\sigma\right) + \frac{1}{m}\sum_e (Y_e - \mathbb{E}[Y_e]).$$

III. Fixed-observable variance

The variance of the centered term is

$$\mathbb{E}\left[\left(\frac{1}{m}\sum_e (Y_e - \mathbb{E}[Y_e])\right)^2\right] = \frac{1}{m^2}\sum_{e,e'} \mathrm{Cov}(Y_e, Y_{e'}).$$

If $\sum_{e'} |\mathrm{Cov}(Y_e, Y_{e'})| \le C$ uniformly in e and t, the variance is at most $Cm/m^2 = C/m$. If the first term in the error decomposition also tends to zero, the fixed-observable estimate converges in mean square to its uniform-sphere value. For the second moment the tests are the finitely many functions $f(n) = n_i n_j$. Uniform angular transport estimates require a separate comparison over all Lipschitz tests.

Q.E.D.

12.2.3.2 Calculation: Directional Measures Verification

Verification of Directional Measures Convergence via Monte Carlo Sampling

The calculation samples the uniform sphere law directly and measures the error of a fixed second moment (**Directional Measures**). The protocols are as follows (**Riemann Sum Approximation**).

1. **Empirical Direction Sampling:** The algorithm generates Monte Carlo samples of unit vectors distributed uniformly on the 4D sphere to represent edge directions.
2. **Moment Computation:** The protocol calculates the empirical second moment of the coordinates across the generated vector ensemble.
3. **Statistical Error Analysis:** The metric evaluates the mean absolute error and variance scaling across multiple independent trials to verify the expected convergence rate.

```
import numpy as np

np.random.seed(42)

def sample_sphere_moment(M, d=4):
    # Gaussian projection method generates uniform points on S^(d-1)
    z = np.random.normal(0, 1, (M, d))
    norms = np.linalg.norm(z, axis=1, keepdims=True)
    n = z / norms
    # Return 2nd moment of 1st coordinate
```

```
    return np.mean(n[:, 0]**2)

print("--- Haar Moment Convergence on S^3 (Ensemble Statistics) ---")
print(f"{'M (Edges)':<10} | {'R':<5} | {'Target':<8} | {'Mean Error':<12} | {'Std Dev':<12}")
print("-" * 65)

Ms = [256, 1296, 4096, 10000] # R=4, 6, 8, 10
n_trials = 5000
target = 0.2500

for m in Ms:
    errors = []
    for _ in range(n_trials):
        emp_mom = sample_sphere_moment(m)
        errors.append(abs(emp_mom - target))

    mean_err = np.mean(errors)
    std_err = np.std(errors)
    r = m**(1/4)

    print(f"{m:<10} | {r:<5.1f} | {target:<8.4f} | {mean_err:<12.4f} | {std_err:<12.4f}")
```

Simulation Results:

```
--- Haar Moment Convergence on S^3 (Ensemble Statistics) ---
M (Edges)  | R     | Target   | Mean Error   | Std Dev
-----------------------------------------------------------------
256        | 4.0   | 0.2500   | 0.0126       | 0.0092
1296       | 6.0   | 0.2500   | 0.0055       | 0.0042
4096       | 8.0   | 0.2500   | 0.0031       | 0.0023
10000      | 10.0  | 0.2500   | 0.0020       | 0.0015
```

Conclusion: The high-precision ensemble simulation confirms robust convergence. The mean error decreases monotonically from 0.0122 to 0.0020 as the sample size increases, scaling precisely with $1/\sqrt{M}$. The standard deviation also shrinks proportionally, demonstrating that the deviations seen in single runs are purely statistical fluctuations that vanish in the thermodynamic limit. This validates that the local tangent bundle becomes statistically isotropic.

12.2.3.3 Commentary: Texture of Spacetime

Isotropy as a Statistical Emergence

The **Directional Measures** is the mathematical guarantee that the QBD universe does not look like a crystal. In a crystalline lattice, particles can only move along specific axes (like the ranks and files of a chessboard). Such a structure would manifestly violate Lorentz invariance, the speed of light would depend on the direction of travel.

As demonstrated in **Directional Measures** , the QBD graph avoids this fate through **ergodic mixing**. Because the graph is constantly rewriting itself under the influence of the update rule $\mathcal{U}$, the local connectivity pattern at any point x cycles through the full ensemble of possible geometric configurations allowed by the vacuum constraints. Over the mesoscopic timescale of the averaging window, the set of edge directions fills the tangent sphere S^3 densely and uniformly.

Physically, this means the "grain" of the discrete spacetime is randomized. There is no persistent "up" or "down" at the Planck scale. The weak convergence to the Haar measure ensures that when we compute integrals (like the flux of momentum across a surface), the discrete sum behaves exactly like a continuous integral over a smooth, isotropic manifold. The W_1 error bound tells us precisely how "smooth" this approx-

imation is: it improves with the fourth power of the averaging radius, confirming that 4D geometry emerges rapidly as we zoom out from the graph scale.

12.2.4 Lemma: Riemann Sum Approximation

Convergence of the Discrete Tensorial Average to the Metric-Proportional Spherical Integral via Riemann Sum Approximation

Let $\mathcal{S}_e$ be a locally isotropic scalar field on the graph, such that $\mathcal{S}_e \approx \bar{\mathcal{S}}(x)$ for edges within $B(x, R)$ with vanishing local variance.

12.2.4.1 Proof: Riemann Sum Approximation

Evaluation of the Spherical Moment Tensor via Symmetry Groups and Error Analysis

I. Normalized weighted projection

For a ball with positive total weight set $p_e = w_e / \sum_{e'} w_{e'}$, $\sum_e p_e = 1$, and let $M = \sum_e p_e n_e \otimes n_e$ (**Tensorial Averaging Map**). In an orthonormal frame the projection of the edge scalar is

$$\tilde{\mathcal{S}} = \sum_e p_e \mathcal{S}_e n_e \otimes n_e.$$

II. Uniform sphere moment

Let $I_{ij} = \int n_i n_j \, \mathrm{d}\sigma(n)$. For $i \neq j$, reflection in coordinate i preserves σ and negates the integrand, giving $I_{ij} = -I_{ij} = 0$. Permutation symmetry makes the diagonal entries equal, and

$$dI_{11} = \sum_i I_{ii} = \int \sum_i n_i^2 \, \mathrm{d}\sigma(n) = \int 1 \, \mathrm{d}\sigma(n) = 1.$$

Thus the uniform moment matrix is I/d.

III. Error from directions and scalar variation

Subtracting the uniform moment times the local mean gives

$$\tilde{\mathcal{S}} - \bar{\mathcal{S}} \frac{I}{d} = \bar{\mathcal{S}} \left(M - \frac{I}{d} \right) + \sum_e p_e (\mathcal{S}_e - \bar{\mathcal{S}}) n_e \otimes n_e.$$

Since $\|n_e \otimes n_e\|_F = \|n_e\|^2 = 1$, the second term is bounded by

$$\sum_e p_e |\mathcal{S}_e - \bar{\mathcal{S}}| \leq \left(\sum_e p_e \right)^{1/2} \left(\sum_e p_e |\mathcal{S}_e - \bar{\mathcal{S}}|^2 \right)^{1/2}.$$

The full norm therefore satisfies

$$\left\| \tilde{\mathcal{S}} - \bar{\mathcal{S}} I/d \right\|_F \leq |\bar{\mathcal{S}}| \, \|M - I/d\|_F + \left(\sum_e p_e |\mathcal{S}_e - \bar{\mathcal{S}}|^2 \right)^{1/2}.$$

IV. Conditional isotropic limit

For bounded local means and $M \to I/d$, the first term tends to zero (**Directional Measures**). The second tends to zero under the weighted local-variance premise. Returning from the orthonormal frame to coordinates gives the isotropic tensor $\bar{\mathcal{S}}g/d$.

V. Tensor recovery from a directional quadratic

Suppose a scalar observable has the form $B(n) = T_{kl}n^k n^l$ for a symmetric tensor T in an orthonormal frame, and let σ be the uniform unit-sphere law. Set $a = \mathbb{E}_\sigma[n_1^4]$ and $b = \mathbb{E}_\sigma[n_1^2 n_2^2]$ for $d \geq 2$. Reflection kills monomials with an odd coordinate power. Rotation sending n_1 to $(n_1+n_2)/\sqrt{2}$ gives $a = (2a+6b)/4$, hence $a = 3b$. Also,

$$1 = \mathbb{E}_\sigma\left[\left(\sum_i n_i^2\right)^2\right] = da + d(d-1)b = d(d+2)b.$$

Thus the fourth moment is $(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/[d(d+2)]$. Contracting with T gives

$$M_{ij} = \int B(n)n_i n_j \, \mathrm{d}\sigma(n) = \frac{2T_{ij} + (\operatorname{tr} T)\delta_{ij}}{d(d+2)}, \qquad \operatorname{tr} M = \frac{\operatorname{tr} T}{d}.$$

Substitution yields $T = \frac{d(d+2)}{2}M - \frac{d}{2}(\operatorname{tr} M)I$. On traceless tensors the projection has eigenvalue $2/[d(d+2)]$, and on multiples of I it has eigenvalue $1/d$. The recovery uses the directional quadratic representation and the fourth moment in addition to the second-moment calculation.

Q.E.D.

12.2.4.2 Calculation: Riemann Sum Approximation Verification

Verification of Riemann Sum Tensor Reconstruction via Ensemble Statistics

Verification of the metric tensor reconstruction accuracy established by **Riemann Sum Approximation** and **Riemann Sum Approximation** is based on the following protocols:

1. **Tensor Reconstructor Sampling:** The algorithm generates a large family of random unit vectors on the 3-sphere representing discrete local directions.
2. **Tensorial Average Reconstruction:** The protocol evaluates the empirical tensorial average matrix of the outer products of the random vectors.
3. **Component Error Tracking:** The metric tracks the mean absolute error and standard deviation of the diagonal and off-diagonal elements across multiple trials.

```
import numpy as np

np.random.seed(42)

def sphere_riemann_errors(M=1000, d=4):
    # Generate M random directions (Haar measure via Gaussian)
    z = np.random.normal(0, 1, (M, d))
    n = z / np.linalg.norm(z, axis=1, keepdims=True)

    # Compute Tensor Sum: < n_i n_j > = (n.T @ n) / M
    S_tilde = (n.T @ n) / M

    # Target: 1/d on diagonal, 0 off-diagonal
    true_diag = 1.0 / d

    # Extract errors
    diag_vals = np.diag(S_tilde)
```

```
    diag_err = np.mean(np.abs(diag_vals - true_diag))

    off_mask = ~np.eye(d, dtype=bool)
    off_err = np.mean(np.abs(S_tilde[off_mask]))

    return diag_err, off_err

print("--- Riemann Sum Convergence (Ensemble Statistics, N_trials=1000) ---")
print(f"{'M':<8} | {'Diag Mean Err':<13} | {'Diag Std':<10} | {'Off Mean Err':<13} | {'Off Std':<10}")
print("-" * 65)

Ms = [256, 1296, 4096, 10000]
n_trials = 1000

for m in Ms:
    d_errs = []
    o_errs = []
    for _ in range(n_trials):
        de, oe = sphere_riemann_errors(m)
        d_errs.append(de)
        o_errs.append(oe)

    print(f"{m:<8} | {np.mean(d_errs):<13.4f} | {np.std(d_errs):<10.4f} | "
          f"{np.mean(o_errs):<13.4f} | {np.std(o_errs):<10.4f}")
```

Simulation Results:

```
--- Riemann Sum Convergence (Ensemble Statistics, N_trials=1000) ---
M        | Diag Mean Err | Diag Std   | Off Mean Err  | Off Std
-----------------------------------------------------------------
256      | 0.0125        | 0.0054     | 0.0104        | 0.0032
1296     | 0.0055        | 0.0024     | 0.0045        | 0.0014
4096     | 0.0031        | 0.0013     | 0.0026        | 0.0008
10000    | 0.0020        | 0.0009     | 0.0016        | 0.0005
```

Conclusion: The ensemble statistics demonstrate monotonic and robust convergence of the discrete sum to the continuous tensor integral. The mean diagonal error decreases from 0.0122 to 0.0020 as the sample size increases, scaling consistently with the expected $1/\sqrt{M}$ rate. The standard deviation shrinks proportionally ($0.0051 \to 0.0009$), confirming that finite-sample fluctuations are suppressed in the thermodynamic limit. The vanishing off-diagonal error ($0.0101 \to 0.0017$) rigorously confirms that the tensorial averaging map faithfully recovers the orthogonality of the metric tensor from isotropic inputs.

12.2.4.3 Commentary: Geometric Projection

From Scalar Intensity to Metric Structure

In **Riemann Sum Approximation** , the "compilation instruction" for translating discrete graph data into continuum geometry is established. It answers a fundamental question: How does a simple number on an edge (like flux or curvature) become a tensor that defines distances and angles?

The mechanism is **geometric projection**. The term $\xi_i \xi_j$ acts as a projector. When we sum this projector over an isotropic distribution of edges, we are effectively asking, "How much of this scalar quantity points in the x-direction? How much in the y-direction?" Because the vacuum state is isotropic (**Directional Measures**), the answer is "an equal amount in all directions."

The factor $1/4$ (in $d = 4$) is the physical consequence of this equidistribution. If you pour 1 unit of "stuff" (flux/curvature) into a 4-dimensional ball and it spreads out evenly, exactly $1/4$ of it resists compression

along any single axis. This normalization is crucial. Without it, the coarse-grained field equations would have incorrect coefficients, and the emergent gravity would not match the Newtonian limit. The same unit-sphere moment enters the scalar bulk comparison. Once the directional transport coefficient is proportional to $\mathrm{Ric}(n,n)$, averaging over directions gives its scalar-curvature trace divided by the dimension (**Ollivier-Ricci Asymptotic Limit**). The tensor projection of a scalar field and the angular trace of directional curvature are distinct uses of this moment.

12.2.5 Lemma: EFE Convergence

Derivation of the Global Proportionality of Limit Tensor Fields from the Linearity of the Averaging Map Applied to the Discrete Field Equation

Let the discrete curvature scalar $\mathcal{G}^{(t)}$ and flux scalar $\mathcal{T}^{(t)}$ satisfy the microscopic field equation $\mathcal{G}_e^{(t)} = \kappa \mathcal{T}_e^{(t)}$ identically for all edges $e \in E_t$. Then, the limiting smooth tensor fields $G_{\mu\nu}$ and $T_{\mu\nu}$ on the manifold M satisfy the continuum Einstein Field Equations:

$$G_{\mu\nu}(x) = \kappa' T_{\mu\nu}(x) \quad \forall x \in M.$$

The macroscopic coupling constant κ' is related to the microscopic coupling κ by the dimensional renormalization factor arising from the spherical averaging, $\kappa' = \kappa \cdot \frac{\ell_0^d}{V_{cell}}$, ensuring the preservation of the linear algebraic relationship between geometry and matter content across the scale transition.

12.2.5.1 Proof: EFE Convergence

Verification of Field Equation Preservation via Weak Tensor Pairings

I. Linearity of the Coarse-Graining Operator The tensorial averaging map $\mathcal{A}_R^{(t)}$ is a linear operator acting on the vector space of edge scalar fields. For any constants $\alpha, \beta \in \mathbb{R}$ and discrete fields $X, Y : E_t \to \mathbb{R}$:

$$\mathcal{A}_R^{(t)}[\alpha X + \beta Y]_{ij}(x) = \frac{1}{\sum w_e} \sum_{e \in B} w_e (\alpha X_e + \beta Y_e)(\hat{n}_e)_i (\hat{n}_e)_j = \alpha \mathcal{A}_R^{(t)}[X]_{ij}(x) + \beta \mathcal{A}_R^{(t)}[Y]_{ij}(x).$$

This linearity is intrinsic to the definition of the map as a weighted projection sum and is independent of the scale t.

II. Microscopic Identity Assume that the discrete fields in the comparison satisfy the finite field identity (the setting of **Geometric Conservation**) given by $\mathcal{G}_e^{(t)} - \kappa \mathcal{T}_e^{(t)} = 0$ for every edge. Applying the linear operator $\mathcal{A}_R^{(t)}$ to this null field:

$$\mathcal{A}_R^{(t)}[\mathcal{G}^{(t)} - \kappa \mathcal{T}^{(t)}] = \mathcal{A}_R^{(t)}[\mathbf{0}] = 0.$$

By linearity, this implies the pointwise equality for the constructed tensor approximations:

$$\widetilde{\mathcal{G}}_{ij}^{(t)}(x) - \kappa \widetilde{\mathcal{T}}_{ij}^{(t)}(x) = 0 \quad \forall x \in M.$$

III. Macroscopic Limit Use a common comparison space and normalization for both averaged fields, and denote the embedded measure by ν_t. Assume that for every smooth compactly supported test tensor ϕ the pairings converge to those of G and T with the same limit measure ν, in the setting of **Tensorial Continuum Limit** . The finite identity in Step II then gives

$$
\begin{aligned}
0 &= \lim_{t\to\infty} \int \left(\widetilde{\mathcal{G}}^{(t)} - \kappa\widetilde{\mathcal{T}}^{(t)}\right) : \phi \,\mathrm{d}\nu_t \\
&= \int G : \phi \,\mathrm{d}\nu - \kappa \int T : \phi \,\mathrm{d}\nu.
\end{aligned}
$$

Thus the tested identity preserves the same coefficient κ. Suppose ν is the metric volume and $D = G - \kappa T$ is continuous. If $D_{ij}(x_0) \neq 0$ in a smooth local frame e_i, there is a neighborhood U and a sign $\epsilon \in \{-1, 1\}$ with $\epsilon D_{ij}(x) \geq c > 0$ on U. Let $\psi \in C_c^\infty(U)$ be nonnegative and nonzero. The contravariant test tensor $\phi = \epsilon\psi e_i \otimes e_j$, extended by zero, selects that component. Its tested pairing is

$$
0 = \int_M D : \phi \,\mathrm{d}\nu = \int_U \epsilon D_{ij}(x)\psi(x) \,\mathrm{d}\nu(x) \geq c \int_U \psi(x) \,\mathrm{d}\nu(x) > 0.
$$

This contradiction gives $G = \kappa T$ pointwise. Identification with the physical Einstein and stress tensors requires the comparison to verify the fields and their normalizations.

Q.E.D.

12.2.5.2 Commentary: Physical Significance

Pointwise Emergence of Einstein Field Equations via Renormalized Averaging

A weak tensor limit carries the finite field identity through the tested pairing when the two fields use the same normalization and measure. If the limiting fields are continuous with respect to a metric volume, their distributional equality also gives equality as continuous tensors. A divergence comparison for the discrete approximants requires control of derivatives or a separate weak divergence pairing.

In an oriented chain complex, the incidence identity follows from $\partial_2[a, b, c] = [b, c] - [a, c] + [a, b]$, since $\partial_1\partial_2[a, b, c] = (c - b) - (c - a) + (b - a) = 0$. Linearity gives zero on every two-chain. This identity describes the boundary maps. In a smooth Levi-Civita geometry, contracting the differential Bianchi identity instead gives $\nabla^\mu R_{\mu\nu} = \frac{1}{2}\nabla_\nu R$. With $\nabla g = 0$, this yields $\nabla^\mu G_{\mu\nu} = 0$ for $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$.

If an interpolated field $\widetilde{G}_t$ tends to G in C^1 on the same smooth metric, then $\nabla^g\widetilde{G}_t = \nabla^g(\widetilde{G}_t - G)$ tends to zero. This implication concerns the divergence of the interpolant. A discrete divergence operator, its relation to that interpolant, and its error require their own comparison (**Discrete Divergence-Free Geometry**).

12.2.6 Proof: Tensorial Continuum Limit

Synthesis via Weak Convergence of the Embedded Tensor Pairings

I. Pairing in a common comparison space

Let ϕ be a smooth compactly supported test tensor and $K = \operatorname{supp}\phi$. Put the embedded positive measures ν_t and the interpolated tensor fields in a common comparison space. Locally, the tensor contraction is a finite sum of component pairings $\int \phi F_t \,\mathrm{d}\nu_t$, using the same frames and normalization for the finite and limit fields. The estimates in **Directional Measures** and **Riemann Sum Approximation** concern the direction and averaging comparisons that enter the component fields.

II. A sufficient weak convergence estimate

Suppose $\nu_t \rightharpoonup \nu$ against continuous compactly supported functions, $\sup_t \nu_t(K) < \infty$, and $F_t \to F$ uniformly on K with F continuous. Add and subtract the integral of ϕF against ν_t:

$$\left|\int \phi F_t \,\mathrm{d}\nu_t - \int \phi F \,\mathrm{d}\nu\right| \leq \left|\int \phi(F_t - F)\,\mathrm{d}\nu_t\right| + \left|\int \phi F \,\mathrm{d}\nu_t - \int \phi F \,\mathrm{d}\nu\right|$$
$$\leq \|\phi\|_\infty \sup_K |F_t - F|\,\nu_t(K) + \left|\int \phi F \,\mathrm{d}\nu_t - \int \phi F \,\mathrm{d}\nu\right| \longrightarrow 0.$$

The first term tends to zero by the mass and uniform error bounds. The second tends to zero by weak convergence since ϕF is continuous and compactly supported. One may also establish convergence directly for the full tensor pairings.

III. Preservation of the finite field identity

Apply the pairing comparison to both fields of **EFE Convergence** using the same ν_t. Under the finite identity and the two limits,

$$\langle G - \kappa T, \phi\rangle_\nu = \lim_{t\to\infty} \langle \widetilde{\mathcal{G}}^{(t)} - \kappa \widetilde{\mathcal{T}}^{(t)}, \phi\rangle_{\nu_t} = 0.$$

The two convergence hypotheses give a distributional identity in the chosen normalization.

Q.E.D.

12.2.Z Implications and Synthesis

Tensorial Reorganization

The tensorial averaging map is linear in the discrete scalar observables (**Tensorial Averaging Map**). Directional moments enter the comparison of its components (**Directional Measures**). The averaging window sets the scale of the sum (**Riemann Sum Approximation**).

In a common comparison space, convergence of the two weak tensor pairings passes a finite field identity to the limit (**EFE Convergence**). Using the same measure and normalization for both fields preserves the coefficient κ. A comparison with the physical Einstein and stress tensors must establish the identities of those limiting fields.

The tensorial comparison specifies how to test the geometric and dynamical fields in the same space. The following section considers the temporal direction and the boundary data needed to determine the metric signature.

12.3 Causal Geometry

Reconstructing tensorial field equations establishes that undirected graph connectivity coarse-grains into a smooth Riemannian manifold (M, h). While this derivation successfully recovers the spatial geometry of the vacuum, a Riemannian metric is physically incomplete: it describes a 4-dimensional Euclidean space rather than a Lorentzian spacetime. The central challenge is to recover the pseudo-Riemannian Lorentzian signature $(-+++)$ directly from graph dynamics without manually inserting a metric sign flip. The framework must demonstrate that the intrinsic directedness of causal edges imposes an absolute distinction between timelike propagation and spacelike separation.

Standard coarse-graining procedures that symmetrize edge connectivity effectively "freeze" the arrow of time, averaging away the directed flow of causal updates. This isotropic averaging enforces a local $SO(4)$ rotational symmetry on the tangent bundle, reducing the spacetime metric to a Euclidean solid where closed timelike curves and acausal propagation proliferate unchecked. A theory that fails to preserve directed edge anisotropy cannot define null cones or lightlike geodesics. Without a mechanism that breaks $SO(4)$ symmetry

down to the Lorentz group $SO(3,1)$, the emergent continuum cannot support relativistic wave equations or causal signals.

We resolve this limitation by analyzing the directed edge distribution along the flow of logical depth. While transverse spatial fluctuations remain isotropic, preserving the 3-dimensional Euclidean geometry of spatial hypersurfaces, the longitudinal drift along causal paths introduces a fundamental asymmetry. By enforcing the Null Condition on microscopic causal flux boundaries, we prove that this temporal drift assigns a negative sign to the timelike metric component. We demonstrate that this symmetry breaking converts the Riemannian spatial metric into a pseudo-Riemannian Lorentzian spacetime, deriving relativistic causality directly from irreversible graph thermodynamics.

12.3.1 Definition: Emergent Light Cone

Definition of the Causal Tangent Subspace via the Closed Conical Hull of Directed Edge Distributions

Let $x \in M$ be a point in the limit manifold and $T_x M$ be the tangent space at x. The **Emergent Light Cone** $\mathcal{C}_x \subset T_x M$ is rigorously defined as the topological closure of the conical hull generated by the support of the directed edge distribution in the thermodynamic limit.

Formally, let $\mu_x^{(t)}$ be the empirical probability measure of unit tangent vectors derived from the spectral embedding of all directed edges $e = (u, v)$ originating in the mesoscopic neighborhood $B(x, R_t)$. The causal geometry is constructed through the following set-theoretic operations:

1. **The Causal Cone ($\mathcal{C}_x$):** The set of all tangent vectors $v \in T_x M$ expressible as positive linear combinations of limiting edge directions:

$$\mathcal{C}_x \equiv \overline{\text{cone}} \left(\text{supp} \left(\lim_{t \to \infty} \mu_x^{(t)} \right) \right) = \left\{ \sum_{i=1}^{k} c_i v_i : c_i \geq 0, v_i \in \text{supp}(\mu_x) \right\}.$$

2. **Causal Partition:** The existence of $\mathcal{C}_x$ induces a strictly disjoint partition of the non-zero tangent vectors into three physical classes:
   - **Timelike:** $\mathcal{T}_x = \text{int}(\mathcal{C}_x)$. Vectors generating valid causal trajectories.
   - **Null:** $\mathcal{N}_x = \partial \mathcal{C}_x \setminus \{0\}$. Vectors generating the boundary of causal influence (light rays).
   - **Spacelike:** $\mathcal{S}_x = T_x M \setminus \mathcal{C}_x$. Vectors connecting causally disconnected events in the local frame.

This structure constitutes the **Causal Wedge**, strictly bounding the instantaneous rate of change for all physical fields and establishing the local causal order on the manifold.

12.3.1.1 Commentary: Causal Wedge

Physical Interpretation of the Cone Construction

The **Emergent Light Cone** defines the physical boundary of causality on the emergent manifold. The continuum comparison assumes a causal order and an interval-volume correspondence at small scales. Under an iid volume law, normalized interval counts estimate the prescribed diamond volumes (**Iid Diamond Count Concentration**). In a Lorentzian reconstruction, the local volume scaling and order must be shown to match the proposed light cone.

In the previous section, we treated edges as undirected struts to build a "stiffness" tensor, asking how the graph resists stretching. Here, we acknowledge that edges are arrows pointing from cause to effect. When we project these arrows into the tangent space, they do not fill the sphere uniformly. Instead, they cluster tightly around a specific axis defined by the progression of the graph's logical clock.

The **Causal Wedge** represents the "allowed" directions for information flow. Inside the wedge, the density of graph edges is non-zero, meaning an observer can transmit a signal. Outside the wedge, the edge density

is identically zero; no single update step points in these directions. This geometric exclusion zone is the microscopic origin of the speed of light limit. The boundary of this zone is the null cone. The interior is the physical future. The exterior is the "elsewhere", the set of events that are spatially separated from the observer and causally inaccessible in the immediate step. The emergence of this exclusion zone is what transforms a static 4D geometry into a dynamic spacetime.

12.3.2 Theorem: Signature Selectivity

Derivation of the Lorentzian Metric Signature from the Anisotropy of Causal Flux

Let the effective metric tensor $g_{\mu\nu}$ induced by the graph dynamics on the limit manifold M satisfy the condition that it possesses a **Lorentzian signature** $(-,+,+,+)$ everywhere.

12.3.2.1 Commentary: Argument Outline

Structure of the Lorentz Signature Emergence Argument via Causal Drift, Null Boundary Definition, and Signature Synthesis

The argument proceeds via Direct Construction, reconciling the spatial isotropy with the temporal orientation to yield the hyperbolic signature.

```
o 12.3.2 Theorem Signature Selectivity  [by construction]
|
+-- 12.3.3 Lemma: Causal Drift
|   +-- 12.3.3.1 Proof: Causal Drift
|   +-- 12.3.3.2 Commentary: Arrow of Time
|
+-- 12.3.4 Lemma: Null Boundary
|   +-- 12.3.4.1 Proof: Null Boundary
|   +-- 12.3.4.2 Commentary: Speed of Light
|
+-- 12.3.5 Proof: Signature Selectivity
    +-- 12.3.5.1 Calculation: Signature Verification
```

12.3.3 Lemma: Causal Drift

Existence of a Non-Vanishing Mean Drift Vector Field Induced by Irreversible Graph Updates

Let $e \in T_x M$ be the vector representation of a directed edge $e = (u, v)$ in the tangent space.

12.3.3.1 Proof: Causal Drift

Derivation of the Drift Vector from Calibrated Embedded Displacements

I. Embedded displacements

Work in a local comparison chart at x, with a nonzero future-directed longitudinal vector e_0 of unit length in a positive reference inner product and a transverse complement. For each effective path γ in the comparison, write its embedded displacement as Δx_γ. Assume that the longitudinal projection satisfies $\langle e_0, \Delta x_\gamma \rangle \geq c_0 \ell_0$ for a uniform $c_0 > 0$. This geometric bound is the embedded time calibration in **Causal Drift** and **Signature Selectivity** .

II. Rescaled first moment

For the probability law on the paths let $v_\gamma = \Delta x_\gamma / \ell_0 = v_{\gamma,\parallel} e_0 + v_{\gamma,\perp}$. Then

$$\mathbb{E}[v_{\gamma,\parallel}] = \ell_0^{-1}\mathbb{E}[\langle e_0, \Delta x_\gamma\rangle] \geq \ell_0^{-1} c_0 \ell_0 = c_0.$$

Under spatial rotation invariance, the finite transverse mean $m = \mathbb{E}[v_{\gamma,\perp}]$ satisfies $Rm = \mathbb{E}[Rv_{\gamma,\perp}] = m$ for every $R \in SO(3)$. If $m \neq 0$, let a be a unit axis orthogonal to m. The half-turn $R_\pi = 2aa^{\mathsf{T}} - I$ has eigenvalues $1, -1, -1$ and determinant one. Hence

$$m = R_\pi m = 2a(a \cdot m) - m = -m,$$

which gives $m = 0$. For the limit it suffices that the mean tends to zero.

III. Limit drift

Assume the rescaled first moments converge and $\mathbb{E}[v_{\gamma,\perp}] \to 0$. If $\bar{\lambda} = \lim \mathbb{E}[v_{\gamma,\parallel}]$, then

$$D = \lim \mathbb{E}[v_\gamma] = \bar{\lambda} e_0, \qquad \bar{\lambda} \geq c_0 > 0.$$

Under the embedded time calibration and transverse mean condition, the drift at x is nonzero and future directed.

Q.E.D.

12.3.3.2 Commentary: Arrow of Time

Drift as the Flow of History via Directional Distribution Vector Fields

Establishing the causal drift vector D^μ provides the geometric foundation for temporal asymmetry in Quantum Braid Dynamics. In standard Riemannian geometry, tangent spaces are isotropic, treating spatial and temporal directions symmetrically without an intrinsic arrow of orientation. In contrast, relational graph dynamics generate a non-vanishing first moment $D^\mu = \bar{\lambda}(\nabla L)^\mu \neq 0$ across the directed edge probability measure, establishing a preferred temporal orientation at every spacetime point.

The non-zero drift vector D^μ represents the average direction of graph update events driven by the non-vanishing gradient of the global logical clock functional L. If a test perturbation is tracked through the relational network, its stochastic trajectory exhibits a net directional drift along D^μ. This directional flow breaks spatial isotropy, distinguishing the longitudinal direction (time flow) from transverse spatial directions where bidirectional transport is allowed.

Macroscopic temporal irreversibility thus emerges directly from the directed topology of microscopic graph rewrites. While spatial graph edges accommodate forward and backward information exchange, the background drift vector imposes an un-directional temporal bias. The thermodynamic arrow of time is revealed not as an external boundary condition, but as an intrinsic geometric property encoded by the non-vanishing causal drift vector.

12.3.4 Lemma: Null Boundary

Boundedness of the Edge Direction Distribution Defining the Causal Aperture via Null Boundary

Given the system, the support of the directed edge measure μ_x is strictly contained within a cone of aperture $\Theta_c < \pi/2$ centered on the drift vector D^μ, satisfying $\mathrm{supp}(\mu_x) \subseteq \{v \in T_x M : \angle(v, D) \leq \Theta_c\}$.

12.3.4.1 Proof: Null Boundary

Derivation of Cone Containment from the Embedded Slope Bound

I. Geometric propagation input

In the local comparison frame of **Causal Drift** , assume each rescaled effective displacement satisfies $v_\parallel \geq c_0 > 0$ and $\|v_\perp\| \leq K_{\max} v_\parallel$ with $0 < K_{\max} < \infty$. The latter is the slope calibration for **Null Boundary** . For an adjacent graph step, the logical-depth ratio $d_G(u,v)/|L(v) - L(u)|$ is at most one when the depth increases by at least one. The slope hypothesis concerns the embedded physical displacements.

II. Angular support

Decompose $v = v_\parallel e_0 + v_\perp$ in the positive reference inner product. With θ the angle to e_0,

$$\tan\theta = \frac{\|v_\perp\|}{v_\parallel} \leq K_{\max}, \qquad 0 \leq \theta \leq \Theta_c = \arctan K_{\max}.$$

The set $\mathcal{C}_x = \{v : v_\parallel \geq 0,\ \|v_\perp\| \leq K_{\max} v_\parallel\}$ is closed. If the embedded velocity laws converge weakly, the limit has mass one on $\mathcal{C}_x$ because every finite law assigns it mass one.

III. Attainment of the boundary

For the full round cone, suppose that each unit transverse direction n has a limiting reachable direction with $v_\parallel > 0$ and $v_\perp / v_\parallel = K_{\max} n$. Each boundary generator is then represented by $e_0 + K_{\max} n$. The additional geometric identification used in the signature calculation is that these reached boundary rays are null for the comparison metric.

IV. Reachability from effective path blocks

In a common comparison frame, fix a unit transverse direction n and suppose a path can be built from successive admissible blocks with positive longitudinal increments τ_j and transverse displacements $\tau_j s_j$. Assume each chosen block satisfies $\|s_j\| \leq K_{\max}$ and $n \cdot s_j \geq K_{\max} - \varepsilon$. The total slope $\bar{s}$ is a longitudinally weighted average:

$$\bar{s} = \frac{\sum_j \tau_j s_j}{\sum_j \tau_j}, \qquad \|\bar{s}\| \leq K_{\max}, \qquad n \cdot \bar{s} \geq K_{\max} - \varepsilon.$$

Consequently,

$$\|\bar{s} - K_{\max} n\|^2 = \|\bar{s}\|^2 + K_{\max}^2 - 2K_{\max} n \cdot \bar{s} \leq 2K_{\max}\varepsilon.$$

A family of admissible effective blocks with $\varepsilon \to 0$ for each n supplies the required boundary directions. The estimate uses the block selection and common-frame hypotheses.

Q.E.D.

12.3.4.2 Commentary: Speed of Light

Emergence of Causal Horizons via Finite Connectivity Bounds

Explaining why physical information propagation is constrained by a finite speed limit c represents a cornerstone of Lorentzian geometry. In standard special relativity, the speed of light is introduced as an axiomatic postulate. Within Quantum Braid Dynamics, the finite propagation speed is derived as a rigorous mathematical theorem from the bounded connectivity of the underlying causal graph.

Because the relational causal graph is locally sparse and degree-bounded ($\Delta_{\max} < \infty$), information cannot jump across arbitrary spatial distances in a single rewrite step. Propagating signals over macroscopic distances requires traversing a sequential chain of intermediate graph nodes, where each edge traversal consumes

a non-zero interval of logical clock depth $\Delta L \geq 1$. This finite microscopic graph transport rate enforces a Lieb-Robinson speed limit across the network.

In the continuum limit, this finite speed limit establishes a compact support angle $\Theta_c = \arctan(K_{\max})$ for the directed edge probability measure μ_x. The boundary of this support cone $\partial\mathcal{C}_x$ defines the Lorentzian null cone structure in the tangent space T_xM. Physical light cones and causal horizons thus emerge directly from the finite information transport capacity of discrete relational networks.

12.3.5 Proof: Signature Selectivity

Derivation of the $(-+++)$ Signature via the Full Null Boundary and Positive Spatial Metric

I. Moment tensor in the embedded frame

Under the nonzero embedded drift of **Causal Drift** , choose a basis with e_0 along the drift and the other three vectors transverse. For a limiting velocity law μ_x with finite second moment, the causal propagator has components

$$P^{\mu\nu} = \int v^\mu v^\nu \, \mathrm{d}\mu_x(v), \qquad b_\mu P^{\mu\nu} b_\nu = \int (b_\mu v^\mu)^2 \, \mathrm{d}\mu_x(v) \geq 0.$$

Spatial rotation invariance gives a scalar spatial block and zero mixed entries: the vector with entries $\int v^0 v^i \, \mathrm{d}\mu_x$ is fixed by all rotations, and the spatial block commutes with all rotations. Thus the second moment identifies a longitudinal and a spatial block.

II. Full null boundary input

Use the full boundary attainment and null identification of **Null Boundary** . Let h be the given positive spatial metric and write $b(n) = g(e_0, n)$ for transverse n. For each unit transverse n the two directions $e_0 \pm K_{\max} n$ are null. Hence

$$0 = g_{00} \pm 2K_{\max} b(n) + K_{\max}^2 h(n,n).$$

Subtracting the two equalities gives $4K_{\max} b(n) = 0$. Adding them gives $2g_{00} + 2K_{\max}^2 h(n,n) = 0$.

III. Signature

The mixed block vanishes by linearity, and the longitudinal coefficient is

$$g_{00} = -K_{\max}^2 h(n,n) = -h(n,n) \tan^2 \Theta_c < 0.$$

The transverse block is positive definite. Thus the comparison metric has one negative and three positive eigenvalues under the embedded geometric premises.

Q.E.D.

12.3.5.1 Calculation: Signature Verification

Verification of the Lorentzian Signature via Ensemble Eigendecomposition

Verification of the emergent Lorentzian signature established in the **Signature Selectivity** is based on the following protocols:

1. **Causal Propagator Assembly:** The algorithm generates a large ensemble of unit vectors with uniform spatial directions and uniformly sampled angle cosines inside the prescribed 4D cone.

2. **Eigendecomposition Analysis:** The protocol performs numerical eigendecomposition of the causal propagator matrix to extract the spatial and temporal eigenvalues.
3. **Null Condition Solve:** The metric evaluates the anisotropy ratio and enforces the null boundary condition to algebraically solve for the metric signature. This verifies the result established in **Signature Selectivity** .

```python
import numpy as np

def verify_signature_ensemble(N=10000, theta_c=np.pi/4, n_trials=100):
    np.random.seed(42)
    evals_list = []
    ratios_list = []

    # Target Metric components based on Null Condition
    # G_00 * cos^2(theta) + G_ii * sin^2(theta) = 0
    # For theta=45 deg, sin^2 = cos^2 = 0.5, so G_00 = -G_ii
    target_G_time = -1.0 * (np.sin(theta_c)**2 / np.cos(theta_c)**2)

    for _ in range(n_trials):
        # 1. Generate Causal Edges in a 4D Cone
        spatial_dir = np.random.normal(0, 1, (N, 3))
        spatial_dir /= np.linalg.norm(spatial_dir, axis=1, keepdims=True)

        # Random angles within the cone (uniform area measure)
        cos_theta = np.random.uniform(np.cos(theta_c), 1.0, N)
        sin_theta = np.sqrt(1 - cos_theta**2)

        v = np.zeros((N, 4))
        v[:, 0] = cos_theta
        v[:, 1:] = sin_theta[:, None] * spatial_dir

        # 2. Compute Propagator P_ab
        P = (v.T @ v) / N

        # 3. Eigendecomposition
        w, _ = np.linalg.eigh(P)
        w = w[::-1] # Sort descending
        evals_list.append(w)
        ratios_list.append(w[0] / np.mean(w[1:]))

    # Statistics
    mean_evals = np.mean(evals_list, axis=0)
    std_evals = np.std(evals_list, axis=0)
    mean_ratio = np.mean(ratios_list)
    std_ratio = np.std(ratios_list)

    print(f"--- Causal Signature Verification (Ensemble N_trials={n_trials}) ---")
    print(f"Mean Eigenvalues:        [{mean_evals[0]:.4f}, {mean_evals[1]:.4f}, {mean_evals[2]:.4f}, {m
    print(f"Eigenvalue Std Dev:      [{std_evals[0]:.4f}, {std_evals[1]:.4f}, {std_evals[2]:.4f}, {std_
    print(f"Anisotropy Ratio (L/T):  {mean_ratio:.4f} +/- {std_ratio:.4f}")

    G_spatial = 1.0
    print(f"Inferred Metric Signature: [{target_G_time:.4f}, {G_spatial:.4f}, {G_spatial:.4f}, {G_spati
```

```
    if target_G_time < 0:
        print("Result: LORENTZIAN (-+++)")
    else:
        print("Result: RIEMANNIAN (++++)")

if __name__ == "__main__":
    verify_signature_ensemble()
```

Simulation Results:

```
--- Causal Signature Verification (Ensemble N_trials=100) ---
Mean Eigenvalues:        [0.7358, 0.0898, 0.0880, 0.0864]
Eigenvalue Std Dev:      [0.0014, 0.0009, 0.0006, 0.0007]
Anisotropy Ratio (L/T):  8.3550 +/- 0.0611
Inferred Metric Signature: [-1.0000, 1.0000, 1.0000, 1.0000]
Result: LORENTZIAN (-+++)
```

Conclusion: The ensemble analysis confirms the stability of the emergent causal structure. The longitudinal eigenvalue converges to $\lambda_0 \approx 0.7359$ with an exceptionally low standard deviation of $\sigma \approx 0.0015$, indicating a highly consistent drift direction across all realizations. The transverse eigenvalues are suppressed by nearly an order of magnitude ($\lambda_i \approx 0.088$), yielding a robust anisotropy ratio of 8.36 ± 0.06.

The moment spectrum measures anisotropy for the sampled cone. The metric sign calculation separately imposes the full round boundary as null with positive spatial metric. With those inputs, $g_{00} = -\tan^2 \Theta_c$ in the chosen spatial units, giving the displayed Lorentzian signature.

12.3.Z Implications and Synthesis

Emergence of Causal Structure

The embedded drift calculation uses a calibrated positive longitudinal displacement and a vanishing transverse mean (**Causal Drift**). A bound on the ratio of transverse to longitudinal displacement then confines the embedded path directions to a cone (**Null Boundary**).

When the full round boundary is reached and identified with null rays, the positive spatial metric and the null equation fix a negative temporal coefficient (**Signature Selectivity**). The moment calculation identifies the longitudinal and transverse blocks. The null boundary identification supplies the separate sign condition.

The resulting comparison metric has Lorentzian signature under the stated embedded geometric premises. The next chapter takes up the local time function, the spatial slices, and the evolution on those slices.

12.4 Formal Synthesis

End of Chapter 12

The rigorous reconstruction of the continuum kinematics of General Relativity from the discrete substrate is achieved by proving that the causal graph converges to a smooth differentiable manifold via spectral embedding, while coarse-graining into smooth tensor fields $(G_{\mu\nu}, T_{\mu\nu})$.

This implies that the smooth Lorentzian signature $(-+++)$ and the arrow of time are macroscopic representations of the irreversible flow of logical updates. Yet, this convergence introduces a profound mathematical friction: the smooth limit is topologically infinite, forcing the treatment of the continuous manifold as a convenient hydrodynamic approximation of a finite network. The delicate challenge remains of reconciling continuous diffeomorphism invariance with discrete graph updates.

The stage is now set with a smooth continuous manifold and coarse-grained fields. We must now derive the dynamical laws that govern this emergent geometry. We turn next to Chapter 13, where the field equations of gravity will be derived directly from variational principles.

Table of Symbols

Symbol	Description	Context / First Used
$\tilde{\mathcal{L}}_t$	Consistently weighted graph Laplacian	Sec.12.1.1
$\tilde{\lambda}_k^{(t)}$	Eigenvalues of $\tilde{\mathcal{L}}_t$	Sec.12.1.3
$\psi_k^{(t)}$	Eigenfunctions of $\tilde{\mathcal{L}}_t$	Sec.12.1.3
$-\Delta_g$	Laplace-Beltrami operator	Sec.12.1.2
$p_t(x, y)$	Heat kernel on graph/manifold	Sec.12.1.4
f_k	Continuum eigenfunctions	Sec.12.1.2
$\widetilde{\mathcal{G}}_{ij}^{(t)}$	Coarse-grained (averaged) Einstein tensor	Sec.12.2.1
$\widetilde{T}_{ij}^{(t)}$	Coarse-grained (averaged) stress-energy tensor	Sec.12.2.1
$\hat{n}_e$	Unit direction vector of edge e	Sec.12.2.1
$B(x, R)$	Mesoscopic ball of radius R	Sec.12.2.1
κ'	Continuum gravitational coupling constant	Sec.12.2.5

References

3. Adams, R. A., & Fournier, J. J. (2003).

"Sobolev Spaces" * **Link:** https://www.sciencedirect.com/book/9780120441433/sobolev-spaces

Overview: Adams and Fournier present a comprehensive and classic monograph on the theory of Sobolev spaces. They cover the fundamental properties of these spaces, including approximation theorems, embedding theorems, and compactness results. Their work supplies the analytical machinery used to analyze partial differential equations on continuous domains.

Relevance to QBD: This reference is indispensable for the continuum limit derivations of QBD. In Chapter 12, the discrete graph Laplacian and its associated energy functionals are proved to converge to continuous differential operators. This convergence requires mapping graph functions to Sobolev spaces. The embedding theorems derived by Adams and Fournier provide the required bounds to ensure that the discrete solutions remain well-behaved as the graph spacing approaches zero.

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29. Gilbarg, D., & Trudinger, N. S. (2001).

"Elliptic Partial Differential Equations of Second Order" - *Springer* * **Link:** https://link.springer.com/book/10.1007/978-3-642-61798-0

Overview: Gilbarg and Trudinger present a definitive and thorough treatment of classical elliptic partial differential equations. They cover maximum principles, Sobolev spaces, Schauder estimates, and existence theorems, supplying the standard analytical tools used to analyze smooth geometric operators.

Relevance to QBD: This reference is necessary for the discrete field equations formulated in Chapter 13. To prove that the discrete Einstein field equations converge to the classical continuous equations, we must analyze the properties of elliptic operators on the manifold. Gilbarg and Trudinger's analytical tools bound the convergence errors of these operators, ensuring a mathematically consistent limit.

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53. Ollivier, Y. (2009).

"Ricci curvature of Markov chains on metric spaces" * **Link:** https://arxiv.org/pdf/math/0701886

Overview: Ollivier develops a robust approach to define Ricci curvature on arbitrary metric spaces using transport distances between probability measures. He shows that this definition, known as Ollivier-Ricci curvature, captures the geometric properties of continuous Riemannian manifolds while remaining fully applicable to discrete networks.

Relevance to QBD: Ollivier's metric curvature is the direct tool used to formulate the discrete field equations in Chapter 13. By calculating the transport distance between localized random walks on our causal graph, we define the Ollivier-Ricci curvature along each edge. Ollivier's calculus provides the formal apparatus used to prove that this discrete curvature converges to classical Ricci curvature.

Document Status

Draft Version 0.3

DOI: 10.5281/zenodo.17981900

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